Note: Linear Spin Wave Theory

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In this article, we summarize the linear spin wave theory (LSWT) described in the literature. **[REVIEW: B8: Consider: 'We present a self-contained summary of LSWT, covering HP formalism, BdG diagonalization, dynamical correlations, and topological magnon properties.']**

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TABLE I. Summary of Notation and Symbols in Linear Spin Wave Theory

Spin Hamiltonian and Related Quantities	
$\hat{H}$	Spin Hamiltonian
$J_{ij}^{\alpha\beta}$	Exchange coupling between sites i and j for spin components α and β
h_j^α	External magnetic field at site j along direction α
$E_{\rm cl}$	Classical ground state energy
$\mathbf{S}_j$	Classical spin vector at site j
$\hat{\mathbf{S}}_j$	Quantum spin operator at site j
$\delta\hat{\mathbf{S}}_j$	Quantum fluctuation operator ($\hat{\mathbf{S}}_j = \mathbf{S}_j + \delta\hat{\mathbf{S}}_j$)
$\widetilde{\mathbf{S}}_j$	Spin operator in the local reference frame
Index Conventions and Coordinate Systems	
i, j	Unit cell indices
μ, ν, σ, ρ	Sublattice indices (typically $a, b, c, \dots$)
$I = (i, \mu), J = (j, \nu)$	Composite site indices combining unit cell and sublattice
$\mathbf{r}_i$	Position of unit cell i
$\boldsymbol{\delta}_\mu$	Position of sublattice μ within the unit cell
$\mathbf{R}_I = \mathbf{r}_i + \boldsymbol{\delta}_\mu$	Position of spin at site $I = (i, \mu)$
α, β	Spin component indices (x, y, z or $+, -, 0$)
$\mathbf{R}_j$	Rotation matrix from laboratory frame to local frame at site j
$\hat{\mathbf{e}}_j^\alpha$	Basis vectors of local frame at site j ($\alpha = +, -, 0$)
Boson Operators and Transformations	
$\hat{a}_j, \hat{a}_j^\dagger$	Bosonic annihilation and creation operators
$\hat{n}_j = \hat{a}_j^\dagger \hat{a}_j$	Boson number operator
$\hat{b}_{\mathbf{k}\mu}, \hat{b}_{\mathbf{k}\mu}^\dagger$	Momentum-space bosonic operators for sublattice μ
$\hat{\beta}_{\mathbf{k}\mu}, \hat{\beta}_{\mathbf{k}\mu}^\dagger$	Diagonalized magnon operators (after Bogoliubov transformation)
$\Psi_{\mathbf{k}} = (\hat{\mathbf{b}}_{\mathbf{k}}, \hat{\mathbf{b}}_{-\mathbf{k}}^\dagger)^T$	Nambu spinor in momentum space
$\widetilde{\Psi}_{\mathbf{k}} = (\hat{\boldsymbol{\beta}}_{\mathbf{k}}, \hat{\boldsymbol{\beta}}_{-\mathbf{k}}^\dagger)^T$	Diagonalized Nambu spinor
Matrices and Transformation Operators	
$\mathsf{H}_{\mathbf{k}}$	Hamiltonian matrix in BdG form
$\mathsf{A}_{\mathbf{k}}, \mathsf{B}_{\mathbf{k}}$	Blocks of the Hamiltonian matrix $\mathsf{H}_{\mathbf{k}}$
$\mathsf{T}_{\mathbf{k}}$	Paraunitary transformation matrix for Bogoliubov diagonalization
$\mathsf{E}_{\mathbf{k}}$	Diagonal matrix of magnon energies
$\mathsf{N}_{\mathbf{k}}(t)$	Time-dependent correlation matrix in magnon basis
U^α	Spin projection operator in laboratory frame
$\mathsf{S}_{\mathbf{k}}$	Matrix encoding spin magnitudes and sublattice phase factors
$\mathsf{C}^{\alpha\beta}(\mathbf{k}, t)$	Correlation matrix in laboratory frame
J	Symplectic metric tensor
Physical Quantities	
$E_{\mathbf{k},\mu}$	Magnon energy for band μ at momentum $\mathbf{k}$
$n_{\mathbf{k},\mu} = 1/(e^{\beta E_{\mathbf{k},\mu}} - 1)$	Bose-Einstein distribution function
E_0	Zero-point energy
$\mathcal{C}_{\mu\nu}^{\alpha\beta}(\mathbf{k}, t)$	Sublattice spin-spin correlation function
$\mathcal{S}^{\alpha\beta}(\mathbf{k}, \omega)$	Dynamic structure factor
$\mathcal{A}^{\alpha\beta}(\mathbf{k}, \omega)$	Spectral function
$\Omega_{n,\mathbf{k}}$	Berry curvature of the n-th magnon band
C_n	Chern number of the n-th magnon band

SUMMARY OF NOTATION AND SYMBOLS

We summarize the symbols and notation used in this notes.

[REVIEW: C13: $R_k^\alpha = U^\alpha S_k$ matrix is heavily used in Sec 4 but absent from this table. Also $\varepsilon_{n,k}$ appears in Sec 5.3 but only $E_{k,\mu}$ is listed.]

I. INTRODUCTION TO SPIN WAVE THEORY (SWT)

For the spin wave theory, the spin model of our interest is

$$\hat{H} = \sum_{ij}\sum_{\alpha,\beta} \hat{S}_i^\alpha J_{ij}^{\alpha\beta} \hat{S}_j^\beta - \sum_{j,\alpha} h_j^\alpha \hat{S}_j^\alpha, \tag{1}$$

where $\mathbf{S}_i = (S_i^x, S_i^y, S_i^z)$ is the spin operator at site i, with S_i^α representing the α-direction component, and ij denotes the *link*, not the lattice indices[1]. The first term is called the exchange Hamiltonian, and the second term is the Zeeman term. Note that many Mott insulators can be generally modeled as in Eqn. (1). **[REVIEW: C22: "Eqn." vs "Eq." used inconsistently throughout. Pick one and unify.] [REVIEW: A4: Summation convention (link sum vs independent index sum) changes between sections without restatement. Consider a global convention in the notation section.]**

(i) **Spin-Exchange interactions:**

$$\sum_{ij}\sum_{\alpha,\beta} \hat{S}_i^\alpha J_{ij}^{\alpha\beta} \hat{S}_j^\beta = \sum_{i\neq j} \hat{\mathbf{S}}_i \cdot \mathbf{J}_{ij} \cdot \hat{\mathbf{S}}_j + \sum_i \hat{\mathbf{S}}_i \cdot \mathbf{A}_i \cdot \hat{\mathbf{S}}_i \tag{2}$$

where the exchange energy is composed of inter-site interactions (first term) and single-ion anisotropies (second term). The exchange matrix $\mathbf{J}_{ij}$ can represent various types of interactions including isotropic exchange, Dzyaloshinskii-Moriya interactions, and anisotropic exchanges such as Kitaev interactions. The single-ion anisotropy term, for example $-A\sum_i (S_i^x)^2$, describes easy-axis anisotropy along x.

(ii) **External magnetic fields and g-tensors:**

$$\sum_j \mathbf{h}_j \cdot \hat{\mathbf{S}}_j = \mu_B \sum_j \sum_{\alpha,\beta} B_j^\alpha g_j^{\alpha\beta} \hat{S}_j^\beta \tag{3}$$

where $\mu_B = 0.057883$ meV/T is the Bohr magneton. In magnetic materials, the g-tensor $g_j^{\alpha\beta}$ mediates the interaction between spins and external magnetic fields and can exhibit significant anisotropy due to spin-orbit coupling and crystal field effects, unlike the isotropic g-factor of free electrons ($g_e \approx 2.002$).

A. Spin to Boson Transformations

A spin wave theory for the model [Eqn. (1)] can be formulated by assuming the quantum spin exhibits small fluctuations around its classical spin configuration:

$$\hat{\mathbf{S}}_j = \mathbf{S}_j + \delta\hat{\mathbf{S}}_j, \tag{4}$$

where $\mathbf{S}_j$ is the classical spin and the operator $\delta\hat{\mathbf{S}}_j$ represents the quantum fluctuation. **[REVIEW: B17: 'the quantum fluctuation' → 'quantum fluctuations' or 'the quantum fluctuation operator'.]** Substituting this into the Hamiltonian H, we obtain:

$$\begin{aligned}
\hat{H} =& \sum_{ij} \left(\mathbf{S}_i + \delta\hat{\mathbf{S}}_i\right) \cdot \mathbf{J}_{ij} \cdot \left(\mathbf{S}_j + \delta\hat{\mathbf{S}}_j\right) - \sum_i \mathbf{h}_i \cdot \left(\mathbf{S}_i + \delta\hat{\mathbf{S}}_i\right) \\
=& E_{\rm cl} + \Big[\sum_{ij} \left(\delta\hat{\mathbf{S}}_i \cdot \mathbf{J}_{ij} \cdot \mathbf{S}_j + \mathbf{S}_i \cdot \mathbf{J}_{ij} \cdot \delta\hat{\mathbf{S}}_j\right) - \sum_i \mathbf{h}_i \cdot \delta\hat{\mathbf{S}}_i\Big] + \sum_{ij} \delta\hat{\mathbf{S}}_i \cdot \mathbf{J}_{ij} \cdot \delta\hat{\mathbf{S}}_j \\
=& E_{\rm cl} + \sum_i \Big(\sum_{j\in \mathsf{L}(i)} \mathbf{S}_j \cdot \mathbf{J}_{ji} - \mathbf{h}_i\Big) \cdot \delta\hat{\mathbf{S}}_i + \sum_{i,j} \delta\hat{\mathbf{S}}_i \cdot \mathbf{J}_{ij} \cdot \delta\hat{\mathbf{S}}_j,
\end{aligned}$$

where $E_{\rm cl} = \sum_{i,j} \mathbf{S}_i \cdot \mathbf{J}_{ij} \cdot \mathbf{S}_j - \sum_i \mathbf{h}_i \cdot \mathbf{S}_i$ is the classical energy, and the summation runs over the links, not the lattice indices[2], and $\sum_i \sum_{j\in\mathsf{L}(i)} = \sum_{\langle i,j\rangle}$. In the second line, we use $\delta\hat{\mathbf{S}}_i \cdot \mathbf{J}_{ij} \cdot \mathbf{S}_j = \mathbf{S}_j \cdot \mathbf{J}_{ji} \cdot \delta\hat{\mathbf{S}}_i$, which follows from $J_{ij}^{\alpha\beta} = J_{ji}^{\beta\alpha}$.

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[1] If we count i and j as independent variables, we change the exchange Hamiltonian: $\sum_{ij}\sum_{\alpha,\beta} S_i^\alpha J_{ij}^{\alpha\beta} S_j^\beta \to \frac{1}{2}\sum_{ij}\sum_{\alpha,\beta} S_i^\alpha J_{ij}^{\alpha\beta} S_j^\beta$.

[2] If we count the indices independently, this becomes $\sum_i \left(2\sum_j \mathbf{S}_j \cdot \mathbf{J}_{ji} - \mathbf{h}_i\right) \cdot \delta\hat{\mathbf{S}}_i$.

The fluctuation operator $\delta\hat{\mathbf{S}}_j$ can be further decomposed into components perpendicular and parallel to the classical spin orientation:

$$\delta\hat{\mathbf{S}}_j = \delta\hat{\mathbf{S}}_j^{\perp} + \delta\hat{\mathbf{S}}_j^{\parallel} \tag{5}$$

Formally, this can be done using the *Holstein-Primakoff* transformation [?], which represents the spin operator in the laboratory frame in terms of bosonic operators:

$$\hat{\mathbf{S}}_j = \sqrt{S_j}\left[\left(1-\frac{\hat{n}_j}{2S_j}\right)^{1/2}\hat{a}_j\hat{\mathbf{e}}_j^{-} + \hat{a}_j^{\dagger}\left(1-\frac{\hat{n}_j}{2S_j}\right)^{1/2}\hat{\mathbf{e}}_j^{+}\right] + (S_j - \hat{n}_j)\hat{\mathbf{e}}_j^{0}, \tag{6}$$

where the operators $\hat{a}_i$ and $\hat{a}_i^{\dagger}$ are the bosonic annihilation and creation operators that satisfy the canonical commutation relation $[\hat{a}_i, \hat{a}_j^{\dagger}] = \delta_{ij}$, and $\hat{n}_j \equiv \hat{a}_j^{\dagger}\hat{a}_j$ is the boson number operator. The vectors $\hat{\mathbf{e}}_j^{\alpha}$ ($\alpha = \pm, 0$) define a local frame of reference. In a more conventional Cartesian basis, one defines $\hat{\mathbf{e}}_j^{\pm} = (\hat{\mathbf{x}}_j \pm i\hat{\mathbf{y}}_j)/\sqrt{2}$ and $\hat{\mathbf{e}}_j^{0} = \hat{\mathbf{z}}_j$ with $\hat{\mathbf{e}}_j^{-} \times \hat{\mathbf{e}}_j^{+} = i\hat{\mathbf{e}}_j^{0}$, where $\hat{\mathbf{e}}_j^{0} = \mathbf{S}_j/S_j = \hat{\mathbf{n}}_j$ denotes the classical spin orientation.

The perpendicular component $\delta\hat{\mathbf{S}}_j^{\perp}$ satisfies $\hat{\mathbf{e}}_j^{0}\cdot\delta\hat{\mathbf{S}}_j^{\perp} = 0$ (where $\hat{\mathbf{e}}_j^{0}$ is the unit vector along the classical spin direction) and contains only odd-order bosonic operators:

$$\delta\hat{\mathbf{S}}_j^{\perp} = \sqrt{S_j}\left[\left(1-\frac{\hat{n}_j}{2S_j}\right)^{1/2}\hat{a}_j\hat{\mathbf{e}}_j^{-} + \hat{a}_j^{\dagger}\left(1-\frac{\hat{n}_j}{2S_j}\right)^{1/2}\hat{\mathbf{e}}_j^{+}\right] = \delta\hat{\mathbf{S}}_j^{(1)} + \delta\hat{\mathbf{S}}_j^{(3)} + \cdots \tag{7}$$

The parallel component, $\delta\hat{\mathbf{S}}_j^{\parallel} = \delta\hat{\mathbf{S}}_j^{(2)} = -\hat{n}_j\hat{\mathbf{e}}_j^{0}$, involves the number operator, where $\hat{\mathbf{e}}_j^{0}$ is the unit vector along the classical spin. **[REVIEW: A3: $\hat{\mathbf{e}}_j$ here should be $\hat{\mathbf{e}}_j^{0}$ to match Sec 1.1 notation. The superscript 0 is dropped inconsistently.]**

Based on the above equations, the Hamiltonian can then be rewritten as:

$$\begin{aligned} H =& E_{\rm cl} + \sum_i \Big(\sum_j \mathbf{S}_j\cdot\mathbf{J}_{ji} - \mathbf{h}_i\Big)\cdot\delta\hat{\mathbf{S}}_i + \sum_{i,j}\delta\hat{\mathbf{S}}_i\cdot\mathbf{J}_{ij}\cdot\delta\hat{\mathbf{S}}_j, \\ =& E_{\rm cl} + H_1 + H_2 + H_3 + H_4 + \cdots, \end{aligned} \tag{8}$$

where the terms could be obtained by plugging $\delta\hat{\mathbf{S}}_j^{\perp} = \delta\hat{\mathbf{S}}_j^{(1)} + \delta\hat{\mathbf{S}}_j^{(3)} + \cdots$, and $\delta\hat{\mathbf{S}}_j^{\parallel} = \delta\hat{\mathbf{S}}_j^{(2)}$.

$$\begin{aligned} E_{\rm cl} =& \sum_{ij}\mathbf{S}_i\cdot\mathbf{J}_{ij}\mathbf{S}_j - \sum_i \mathbf{h}_i\cdot\mathbf{S}_i = \sum_{ij}\tilde{J}_{ij}^{00}S_iS_j - \sum_i S_i^0 h_i, \\ H_1 =& \sum_i \Big(\sum_{j\in\mathsf{L}(i)}\mathbf{S}_j\cdot\mathbf{J}_{ji} - \mathbf{h}_i\Big)\cdot\delta\hat{\mathbf{S}}_i^{(1)} = \sum_i \frac{\partial E_{\rm cl}}{\partial\mathbf{S}_i}\cdot\delta\hat{\mathbf{S}}_i^{(1)}, \\ H_2 =& \sum_{ij}\delta\hat{\mathbf{S}}_i^{(1)}\cdot\mathbf{J}_{ij}\cdot\delta\hat{\mathbf{S}}_j^{(1)} + \sum_i\Big(\sum_{j\in\mathsf{L}(i)}\mathbf{S}_j\cdot\mathbf{J}_{ji} - \mathbf{h}_i\Big)\cdot\delta\hat{\mathbf{S}}_i^{(2)}, \\ H_3 =& \sum_{ij}\big(\delta\hat{\mathbf{S}}_i^{(1)}\cdot\mathbf{J}_{ij}\cdot\delta\hat{\mathbf{S}}_j^{(2)} + \delta\hat{\mathbf{S}}_i^{(2)}\cdot\mathbf{J}_{ij}\cdot\delta\hat{\mathbf{S}}_j^{(1)}\big), \quad \text{and so on.} \end{aligned} \tag{9}$$

Note that the linear boson term vanishes when the spin configuration corresponds to a classical extremum:

$$\sum_i \frac{\partial E_{\rm cl}}{\partial\mathbf{S}_i}\cdot\delta\hat{\mathbf{S}}_i^{(1)} = 0. \tag{10}$$

To demonstrate this, consider the Lagrangian with a Lagrange multiplier enforcing the spin magnitude constraint:

$$\mathcal{L} = E_{\rm cl} - \sum_j \frac{\lambda_j}{2}\big(|\mathbf{S}_j|^2 - S_j^2\big). \tag{11}$$

Taking the derivative with respect to $\mathbf{S}_i$, we get:

$$\frac{\partial\mathcal{L}}{\partial\mathbf{S}_i} = \frac{\partial E_{\rm cl}}{\partial\mathbf{S}_i} - \lambda_i\mathbf{S}_i = 0, \tag{12}$$

where $\frac{\partial E_{\rm cl}}{\partial \mathbf{S}_i} = \sum_j \mathbf{S}_j \cdot \mathbf{J}_{ji} - \mathbf{h}_i$. Since $\delta\hat{\mathbf{S}}_i^{\perp}$ is perpendicular to $\mathbf{S}_i$, it follows that:

$$\boxed{\frac{\partial E_{\rm cl}}{\partial \mathbf{S}_i} \cdot \delta\hat{\mathbf{S}}_i^{\perp} = \lambda_i \mathbf{S}_i \cdot \delta\hat{\mathbf{S}}_i^{\perp} = 0, \text{ for all } i.} \tag{13}$$

Thus, the linear boson term, as well as any odd terms related to classical energy differentiation, vanishes when the spin configuration is a classical stable point.

Comment — Alternatively, *Dyson-Maleev* transformation [**? ?**] could map spin operators onto bosonic creation and annihilation operators $(a_i^\dagger, a_i)$ as well:

$$S_i^+ = \sqrt{2S_i}\Big(1 - \frac{a_i^\dagger a_i}{2S_i}\Big)^{\xi} a_i, \quad S_i^- = \sqrt{2S_i}\, a_i^\dagger \Big(1 - \frac{a_i^\dagger a_i}{2S_i}\Big)^{1-\xi}, \quad S_i^z = S_i - a_i^\dagger a_i, \tag{14}$$

where $\xi = 1/2$ for the Holstein-Primakoff transformation, and $\xi = 1$ for the Dyson-Maleev transformation. While both transformations map spin operators onto bosonic operators, the Dyson-Maleev transformation does not preserve Hermiticity. For instance, $(S_i^+)^\dagger$ is not equal to S_i^-:

$$(S_i^+)^\dagger = \sqrt{2S_i} a_i^\dagger \Big(1 - \frac{a_i^\dagger a_i}{2S_i}\Big)^{\xi} = S_i^- \Big(1 - \frac{a_i^\dagger a_i}{2S_i}\Big)^{2\xi - 1}, \tag{15}$$

whereas for $\xi = 1/2$ (the Holstein-Primakoff transformation), Hermiticity of the spin operators is preserved.

B. Rotations for Spin Models

To analyze an ordered spin system using linear spin-wave theory (LSWT), it is useful to rotate the spins at each site from the *laboratory* reference frame $\{\hat{\mathbf{x}}, \hat{\mathbf{y}}, \hat{\mathbf{z}}\}$ to a *local* reference frame $\{\hat{\mathbf{x}}_j, \hat{\mathbf{y}}_j, \hat{\mathbf{z}}_j\}$ (aligned with the classical spin ordering, where $\hat{\mathbf{z}}_j$ coincides with the spin's quantization axis). The transformation is given by:

$$\hat{\mathbf{S}}_j = \mathbf{R}_j\, \widetilde{\mathbf{S}}_j = \widetilde{S}_j^x \hat{\mathbf{x}}_j + \widetilde{S}_j^y \hat{\mathbf{y}}_j + \widetilde{S}_j^z \hat{\mathbf{z}}_j, \tag{16}$$

where $\widetilde{\mathbf{S}}_j$ is the spin operator in the local reference frame at site j. By applying the rotation to all spins in a unit cell, the spin Hamiltonian is transformed from the laboratory frame to the local frame as:

$$\hat{H} = \sum_{ij} \sum_{\alpha,\beta} S_i^\alpha J_{ij}^{\alpha\beta} S_j^\beta - \sum_{j,\alpha} h_j^\alpha S_j^\alpha = \sum_{ij} \sum_{\alpha,\beta} \widetilde{S}_i^\alpha \widetilde{J}_{ij}^{\alpha\beta} \widetilde{S}_j^\beta - \sum_{j,\alpha} \widetilde{h}_j^\alpha \widetilde{S}_j^\alpha, \tag{17}$$

where the rotated couplings are defined as:

$$\widetilde{J}_{ij}^{\alpha\beta} = [\widetilde{\mathbf{J}}_{ij}]^{\alpha\beta} = [\mathbf{R}_i^{\rm T} \mathbf{J}_{ij} \mathbf{R}_j]^{\alpha\beta}, \quad \text{and} \quad \widetilde{h}_j^\alpha = [\widetilde{\mathbf{h}}_j]^\alpha = [\mathbf{h}_j^{\rm T} \mathbf{R}_j]^\alpha. \tag{18}$$

In practice, the rotation from the crystal frame to the local frame is performed in two stages, expressed as $\mathbf{R}_i = \mathbf{R}^\varphi \cdot \mathbf{R}^\theta$. The combined rotation matrix is:

$$\mathbf{R}(\theta, \varphi) = \mathbf{R}^\varphi \cdot \mathbf{R}^\theta = \begin{pmatrix} \cos\theta\cos\varphi & -\sin\varphi & \sin\theta\cos\varphi \\ \cos\theta\sin\varphi & \cos\varphi & \sin\theta\sin\varphi \\ -\sin\theta & 0 & \cos\theta \end{pmatrix}, \tag{19}$$

where θ is the angle relative to the z-axis in lab frame and φ is the azimuthal angle in the xy-plane. Note that the classical spin orientation $\mathbf{S}_j$ satisfies:

$$\mathbf{S}_j = \mathbf{R}_j\, \hat{\mathbf{z}} = \mathbf{R}^{\varphi_j} \mathbf{R}^{\theta_j} \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} = \mathbf{R}^{\varphi_j} \begin{pmatrix} \sin\theta_j \\ 0 \\ \cos\theta_j \end{pmatrix} = \begin{pmatrix} \sin\theta_j \cos\varphi_j \\ \sin\theta_j \sin\varphi_j \\ \cos\theta_j \end{pmatrix}. \tag{20}$$

For later uses, it is also convenient to express the spin operator using ladder operators with complex vector:

$$\hat{\mathbf{S}}_j = \mathbf{R}_j\, \widetilde{\mathbf{S}}_j = \frac{\widetilde{S}_j^+ \hat{\mathbf{e}}_j^- + \widetilde{S}_j^- \hat{\mathbf{e}}_j^+}{\sqrt{2}} + \widetilde{S}_j^0 \hat{\mathbf{e}}_j^0, \tag{21}$$

where the spin operators are $\widetilde{S}_j^{\pm} \equiv \widetilde{S}_j^{x} \pm i\widetilde{S}_j^{y}$, $\widetilde{S}_j^{0} = \widetilde{S}_j^{z}$, and $[\widetilde{S}_j^{z}, \widetilde{S}_j^{\pm}] = \pm\widetilde{S}_j^{\pm}$. The complex vectors are

$$\hat{\mathbf{e}}_j^{\alpha} \equiv \mathbf{R}_j \hat{\mathbf{e}}^{\alpha} : \quad \hat{\mathbf{e}}_j^{\pm} = \mathbf{R}_j \left(\frac{\hat{\mathbf{x}} \pm i\hat{\mathbf{y}}}{\sqrt{2}} \right) \text{ and } \hat{\mathbf{e}}_j^{0} \equiv \mathbf{R}_j \hat{\mathbf{z}}, \tag{22}$$

where $\alpha = \pm, 0, (x, y, z)$ and the $\hat{\mathbf{e}}_j^0$-direction represents the classical magnetic orientation at site j. These complex vectors $\hat{\mathbf{e}}_j^{\alpha}$ introduce a rotated Hamiltonian similar to the previous one:

$$\hat{H} = \sum_{ij} \sum_{\alpha,\beta=0,\pm} \widetilde{S}_i^{\alpha} \widetilde{J}_{ij}^{\bar{\alpha}\bar{\beta}} \widetilde{S}_j^{\beta} - \sum_{j} \sum_{\alpha=0,\pm} \widetilde{h}_j^{\alpha} \widetilde{S}_j^{\alpha}. \tag{23}$$

where $\bar{(\cdot)} = -(\cdot)$, for example $\alpha = \pm, (0)$ and $\bar{\alpha} = \mp, (0)$. The couplings $\widetilde{J}_{ij}^{\bar{\alpha}\bar{\beta}}$ and magnetic fields $\widetilde{h}_j^{\alpha}$ are defined as

$$\widetilde{J}_{ij}^{\alpha\beta} \equiv \hat{\mathbf{e}}_i^{\alpha} \cdot \widetilde{\mathbf{J}}_{ij} \cdot \hat{\mathbf{e}}_j^{\beta} \quad \text{and} \quad \widetilde{h}_j^{\alpha} \equiv \widetilde{\mathbf{h}}_j \cdot \hat{\mathbf{e}}_j^{\alpha}. \tag{24}$$

The transformation from xyz to ± 0 basis is given by:

$$\mathbf{C} = \begin{pmatrix} \hat{\mathbf{e}}^{+} & \hat{\mathbf{e}}^{-} & \hat{\mathbf{e}}^{0} \end{pmatrix} = \begin{pmatrix} [\hat{\mathbf{e}}^{+}]^x & [\hat{\mathbf{e}}^{-}]^x & [\hat{\mathbf{e}}^{0}]^x \\ [\hat{\mathbf{e}}^{+}]^y & [\hat{\mathbf{e}}^{-}]^y & [\hat{\mathbf{e}}^{0}]^y \\ [\hat{\mathbf{e}}^{+}]^z & [\hat{\mathbf{e}}^{-}]^z & [\hat{\mathbf{e}}^{0}]^z \end{pmatrix} = \begin{pmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & 0 \\ \frac{i}{\sqrt{2}} & -\frac{i}{\sqrt{2}} & 0 \\ 0 & 0 & 1 \end{pmatrix} \tag{25}$$

where $[\hat{\mathbf{e}}_i^{\alpha}]^{\beta}$ denotes the $\beta = x, y, z$ component of the basis vector $\hat{\mathbf{e}}_i^{\alpha}$ (with $\alpha = +, -, 0$). These components are obtained from:

$$\mathbf{U}_j = \mathbf{R}_j \mathbf{C} = \begin{pmatrix} [\hat{\mathbf{e}}_j^{+}]^x & [\hat{\mathbf{e}}_j^{-}]^x & [\hat{\mathbf{e}}_j^{0}]^x \\ [\hat{\mathbf{e}}_j^{+}]^y & [\hat{\mathbf{e}}_j^{-}]^y & [\hat{\mathbf{e}}_j^{0}]^y \\ [\hat{\mathbf{e}}_j^{+}]^z & [\hat{\mathbf{e}}_j^{-}]^z & [\hat{\mathbf{e}}_j^{0}]^z \end{pmatrix} = \begin{pmatrix} R_j^{xx} & R_j^{xy} & R_j^{xz} \\ R_j^{yx} & R_j^{yy} & R_j^{yz} \\ R_j^{zx} & R_j^{zy} & R_j^{zz} \end{pmatrix} \begin{pmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & 0 \\ \frac{i}{\sqrt{2}} & -\frac{i}{\sqrt{2}} & 0 \\ 0 & 0 & 1 \end{pmatrix} \tag{26}$$

The transformed coupling tensor $\mathbf{C}^{\dagger}\mathbf{J}\mathbf{C} =$ is expressed as:

$$\begin{pmatrix} J_{ij}^{-+} & J_{ij}^{--} & J_{ij}^{-0} \\ J_{ij}^{++} & J_{ij}^{+-} & J_{ij}^{+0} \\ J_{ij}^{0+} & J_{ij}^{0-} & J_{ij}^{00} \end{pmatrix} = \begin{pmatrix} \frac{1}{\sqrt{2}} & -\frac{i}{\sqrt{2}} & 0 \\ \frac{1}{\sqrt{2}} & \frac{i}{\sqrt{2}} & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} J_{ij}^{xx} & J_{ij}^{xy} & J_{ij}^{xz} \\ J_{ij}^{yx} & J_{ij}^{yy} & J_{ij}^{yz} \\ J_{ij}^{zx} & J_{ij}^{zy} & J_{ij}^{zz} \end{pmatrix} \begin{pmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & 0 \\ \frac{i}{\sqrt{2}} & -\frac{i}{\sqrt{2}} & 0 \\ 0 & 0 & 1 \end{pmatrix} \tag{27}$$

[REVIEW: C23: Remove commented-out explicit form (% begin{pmatrix}...) before final version.]
with elements defined as $\widetilde{J}_{ij}^{\alpha\beta} = \hat{\mathbf{e}}_i^{\alpha} \cdot \widetilde{\mathbf{J}}_{ij} \cdot \hat{\mathbf{e}}_j^{\beta}$:

$$\begin{aligned} J_{ij}^{--} &= \tfrac{1}{2}\big(J_{ij}^{xx} - J_{ij}^{yy} - i(J_{ij}^{xy} + J_{ij}^{yx})\big), & J_{ij}^{++} &= \tfrac{1}{2}\big(J_{ij}^{xx} - J_{ij}^{yy} + i(J_{ij}^{xy} + J_{ij}^{yx})\big), & & \\ J_{ij}^{-+} &= \tfrac{1}{2}\big(J_{ij}^{xx} + J_{ij}^{yy} + i(J_{ij}^{xy} - J_{ij}^{yx})\big), & J_{ij}^{+-} &= \tfrac{1}{2}\big(J_{ij}^{xx} + J_{ij}^{yy} - i(J_{ij}^{xy} - J_{ij}^{yx})\big), & & \\ J_{ij}^{0\pm} &= \tfrac{1}{\sqrt{2}}\big(J_{ij}^{zx} \pm iJ_{ij}^{zy}\big), & J_{ij}^{\pm 0} &= \tfrac{1}{\sqrt{2}}\big(J_{ij}^{xz} \pm iJ_{ij}^{yz}\big), & J_{ij}^{00} &= J_{ij}^{zz}. \end{aligned} \tag{28}$$

C. Bosonic representations for spin Hamiltonian

In this section, we derive the explicit bosonic Hamiltonian for the spin model.

$$\hat{H} = E_{\text{cl}} + \sum_i \Big(\sum_{j \in \mathsf{L}(ij)} \mathbf{S}_j \cdot \mathbf{J}_{ji} - \mathbf{h}_i \Big) \cdot \delta\hat{\mathbf{S}}_i + \sum_{ij \in \mathsf{L}} \delta\hat{\mathbf{S}}_i \cdot \mathbf{J}_{ij} \cdot \delta\hat{\mathbf{S}}_j, \tag{29}$$

where E_{cl} is the classical energy, $\mathbf{S}_j$ represents the spin vector at site j, $\mathbf{J}_{ji}$ is the exchange interaction tensor, $\mathbf{h}_i$ is the external field, and $\delta\hat{\mathbf{S}}_i$ denotes the spin fluctuation operator.

Let us divide the spin model into even and odd bosonic operator terms,

$$H_{\text{even}} = \sum_{ij \in \mathsf{L}} \Big(\delta\hat{\mathbf{S}}_i^{\perp} \cdot \mathbf{J}_{ij} \cdot \delta\hat{\mathbf{S}}_j^{\perp} + \delta\hat{\mathbf{S}}_i^{\parallel} \cdot \mathbf{J}_{ij} \cdot \delta\hat{\mathbf{S}}_j^{\parallel} \Big) + \sum_i \Big(\sum_{j \in \mathsf{L}(ij)} \mathbf{S}_j \cdot \mathbf{J}_{ji} - \mathbf{h}_i \Big) \cdot \delta\hat{\mathbf{S}}_i^{\parallel}, [REVIEW: A8: Last term is \sum h_i \hat{n}_i \text{ but sh} \tag{30a}$$

$$H_{\text{odd}}^{(m \geq 3)} = \sum_{ij \in \mathsf{L}} \Big(\delta\hat{\mathbf{S}}_i^{\perp} \cdot \mathbf{J}_{ij} \cdot \delta\hat{\mathbf{S}}_j^{\parallel} + \delta\hat{\mathbf{S}}_i^{\parallel} \cdot \mathbf{J}_{ij} \cdot \delta\hat{\mathbf{S}}_j^{\perp} \Big). \tag{30b}$$

and substituting the following spin-to-boson transformation:

$$\delta\hat{\mathbf{S}}_i^{\perp} = \frac{\widetilde{S}_i^{+}\hat{\mathbf{e}}_i^{-} + \widetilde{S}_i^{-}\hat{\mathbf{e}}_i^{+}}{\sqrt{2}} = \sqrt{S_i}\left[\left(1-\frac{\hat{a}_i^{\dagger}\hat{a}_i}{2S_i}\right)^{1/2}\hat{a}_i\hat{\mathbf{e}}_i^{-} + \hat{a}_i^{\dagger}\left(1-\frac{\hat{a}_i^{\dagger}\hat{a}_i}{2S_i}\right)^{1/2}\hat{\mathbf{e}}_i^{+}\right], \quad \text{and} \quad \delta\hat{\mathbf{S}}_i^{\parallel} = -\hat{n}_i\hat{\mathbf{e}}_i, \tag{31}$$

[REVIEW: A3: $\hat{\mathbf{e}}_i$ here should be $\hat{\mathbf{e}}_i^0$ for consistency with Sec 1.1 notation.] where $\hat{\mathbf{e}}_i^{\pm}$ are the raising and lowering unit vectors, $\hat{\mathbf{e}}_i$ is the unit vector along the local quantization axis, and $\hat{n}_i = \hat{a}_i^{\dagger}\hat{a}_i$ is the boson number operator. This boson mapping results in the SWT Hamiltonian. While the odd terms can also be obtained easily, we will focus on the even terms as they are of primary interest for the linear spin wave theory.

The even Hamiltonian term, H_{even}, consists of terms involving an even number of bosonic operators:

$$\delta\hat{\mathbf{S}}_i^{\perp}\cdot\mathbf{J}_{ij}\cdot\delta\hat{\mathbf{S}}_j^{\perp} = \frac{1}{2}\left(\widetilde{S}_i^{+}\widetilde{J}_{ij}^{--}\widetilde{S}_j^{+} + \widetilde{S}_i^{+}\widetilde{J}_{ij}^{-+}\widetilde{S}_j^{-} + \widetilde{S}_i^{-}\widetilde{J}_{ij}^{+-}\widetilde{S}_j^{+} + \widetilde{S}_i^{-}\widetilde{J}_{ij}^{++}\widetilde{S}_j^{-}\right), \tag{32}$$

This becomes:

$$\begin{aligned}
\frac{1}{2}\widetilde{S}_i^{+}\widetilde{J}_{ij}^{-+}\widetilde{S}_j^{-} &= \sqrt{S_iS_j}\widetilde{J}_{ij}^{-+}\left(1-\frac{\hat{n}_i}{2S_i}\right)^{1/2}\hat{a}_i\hat{a}_j^{\dagger}\left(1-\frac{\hat{n}_j}{2S_j}\right)^{1/2} = t_{ij}^{-+}\left[\hat{a}_i\hat{a}_j^{\dagger} - \tfrac{1}{4}\left(\tfrac{\hat{n}_i\hat{a}_i\hat{a}_j^{\dagger}}{S_i} + \tfrac{\hat{a}_i\hat{a}_j^{\dagger}\hat{n}_j}{S_j}\right) + \mathcal{O}(S^{-2})\right]\\
\frac{1}{2}\widetilde{S}_i^{-}\widetilde{J}_{ij}^{+-}\widetilde{S}_j^{+} &= \sqrt{S_iS_j}\widetilde{J}_{ij}^{+-}\hat{a}_i^{\dagger}\left(1-\frac{\hat{n}_i}{2S_i}\right)^{1/2}\left(1-\frac{\hat{n}_j}{2S_j}\right)^{1/2}\hat{a}_j = t_{ij}^{+-}\left[\hat{a}_i^{\dagger}\hat{a}_j - \tfrac{1}{4}\left(\tfrac{\hat{a}_i^{\dagger}\hat{n}_i\hat{a}_j}{S_i} + \tfrac{\hat{a}_i^{\dagger}\hat{n}_j\hat{a}_j}{S_j}\right) + \mathcal{O}(S^{-2})\right]\\
\frac{1}{2}\widetilde{S}_i^{+}\widetilde{J}_{ij}^{--}\widetilde{S}_j^{+} &= \sqrt{S_iS_j}\widetilde{J}_{ij}^{--}\left(1-\frac{\hat{n}_i}{2S_i}\right)^{1/2}\hat{a}_i\left(1-\frac{\hat{n}_j}{2S_j}\right)^{1/2}\hat{a}_j = t_{ij}^{--}\left[\hat{a}_i\hat{a}_j - \tfrac{1}{4}\left(\tfrac{\hat{n}_i\hat{a}_i\hat{a}_j}{S_i} + \tfrac{\hat{a}_i\hat{n}_j\hat{a}_j}{S_j}\right) + \mathcal{O}(S^{-2})\right]\\
\frac{1}{2}\widetilde{S}_i^{-}\widetilde{J}_{ij}^{++}\widetilde{S}_j^{-} &= \sqrt{S_iS_j}\widetilde{J}_{ij}^{++}\hat{a}_i^{\dagger}\left(1-\frac{\hat{n}_i}{2S_i}\right)^{1/2}\hat{a}_j^{\dagger}\left(1-\frac{\hat{n}_j}{2S_j}\right)^{1/2} = t_{ij}^{++}\left[\hat{a}_i^{\dagger}\hat{a}_j^{\dagger} - \tfrac{1}{4}\left(\tfrac{\hat{a}_i^{\dagger}\hat{n}_i\hat{a}_j^{\dagger}}{S_i} + \tfrac{\hat{a}_i^{\dagger}\hat{a}_j^{\dagger}\hat{n}_j}{S_j}\right) + \mathcal{O}(S^{-2})\right],
\end{aligned}$$

where we define $t_{ij}^{\alpha\beta} = \sqrt{S_iS_j}\widetilde{J}_{ij}^{\alpha\beta}$. Next, the other terms in Eqn. (30a) are:

$$\begin{aligned}
\sum_i\Big(\sum_{j\in\mathsf{L}(i)}\mathbf{S}_j\cdot\mathbf{J}_{ji} - \mathbf{h}_i\Big)\cdot\delta\hat{\mathbf{S}}_i^{\parallel} &= \sum_i\Big(\widetilde{h}_i^0 - \sum_{j\in\mathsf{L}(i)}S_j\cdot\widetilde{J}_{ji}^{00}\Big)\hat{n}_i,\\
\delta\hat{\mathbf{S}}_i^{\parallel}\cdot\mathbf{J}_{ij}\cdot\delta\hat{\mathbf{S}}_j^{\parallel} &= \hat{n}_i\widetilde{J}_{ij}^{00}\hat{n}_j,
\end{aligned} \tag{33}$$

where $\widetilde{h}_i^0$ is the magnetic field along the local quantization axis and $\widetilde{J}_{ij}^{00}$ represents the longitudinal component of the exchange interaction.

Combining all terms, we can express the quadratic and quartic Hamiltonian contributions, H_2 and H_4, as:

$$H_2 = \sum_{ij}\left[t_{ij}^{--}\hat{a}_i\hat{a}_j + t_{ij}^{-+}\hat{a}_i\hat{a}_j^{\dagger} + t_{ij}^{+-}\hat{a}_i^{\dagger}\hat{a}_j + t_{ij}^{++}\hat{a}_i^{\dagger}\hat{a}_j^{\dagger} - \widetilde{J}_{ij}^{00}(S_i\hat{n}_j + S_j\hat{n}_i)\right] + \sum_i h_i\hat{n}_i, \tag{34}$$

$$H_4 = \sum_{ij}\widetilde{J}_{ij}^{00}\hat{n}_i\hat{n}_j - \left[t_{ij}^{--}\left(\tfrac{\hat{n}_i}{S_i}+\tfrac{\hat{n}_j}{S_j}\right)\hat{a}_i\hat{a}_j + t_{ij}^{++}\hat{a}_i^{\dagger}\hat{a}_j^{\dagger}\left(\tfrac{\hat{n}_i}{S_i}+\tfrac{\hat{n}_j}{S_j}\right) + t_{ij}^{-+}\left(\tfrac{\hat{n}_i\hat{a}_i\hat{a}_j^{\dagger}}{S_i}+\tfrac{\hat{a}_i\hat{a}_j^{\dagger}\hat{n}_j}{S_j}\right) + t_{ij}^{+-}\left(\tfrac{\hat{a}_i^{\dagger}\hat{n}_i\hat{a}_j}{S_i}+\tfrac{\hat{a}_i^{\dagger}\hat{n}_j\hat{a}_j}{S_j}\right)\right]. \tag{35}$$

Note that if the anisotropic couplings are absent in the rotated exchange matrix $\widetilde{J}^{xy} = \widetilde{J}^{yx} = 0$, and $\widetilde{J}^{xx} = \widetilde{J}^{yy}$ are equal, then the anomalous contribution, such as $a^{\dagger}a^{\dagger}$, vanishes, see Eq. (28). Furthermore, when the spins' magnitudes equal $S_i = S$, the expansions

$$H = S^2\tilde{E}_{cl} + S^1\tilde{H}_2 + S^{1/2}\tilde{H}_3 + S^0\tilde{H}_4 + \cdots. \tag{36}$$

D. Momentum Space Representations

[REVIEW: A9: δ_{ij} is used but never defined. δ_μ was defined as sublattice position. Presumably $\delta_{ij} = R_j - R_i$ (bond vector). State explicitly.]

In this section, we shall find the momentum space representation for the quadratic spin wave Hamiltonian (H_2). The quadratic bosonic Hamiltonian we have in Eqn. (34) is

$$\begin{aligned}
H_2 &= \sum_{ij}\left[t_{ij}^{--}\hat{a}_i\hat{a}_j + t_{ij}^{-+}\hat{a}_i\hat{a}_j^{\dagger} + t_{ij}^{+-}\hat{a}_i^{\dagger}\hat{a}_j + t_{ij}^{++}\hat{a}_i^{\dagger}\hat{a}_j^{\dagger} - \widetilde{J}_{ij}^{00}(S_i\hat{a}_j^{\dagger}\hat{a}_j + S_j\hat{a}_i^{\dagger}\hat{a}_i)\right] + \sum_i\widetilde{h}_i^0\hat{a}_i^{\dagger}\hat{a}_i\\
&= \sum_{ij}\left[t_{ij}^{--}\hat{a}_i\hat{a}_j + t_{ij}^{-+}\hat{a}_i\hat{a}_j^{\dagger} + t_{ij}^{+-}\hat{a}_i^{\dagger}\hat{a}_j + t_{ij}^{++}\hat{a}_i^{\dagger}\hat{a}_j^{\dagger}\right] + \sum_i\mu_i\hat{a}_i^{\dagger}\hat{a}_i,
\end{aligned} \tag{37}$$

where the (effective) chemical potential μ_i is

$$\mu_i = \widetilde{h}_i^0 - \sum_{j\in \mathsf{L}(i)} S_j \widetilde{J}_{ij}^{00} = \frac{\partial E_{\rm cl}}{\partial \mathbf{S}_i}\cdot \hat{\mathbf{e}}_i^0. \tag{38}$$

We assume translational invariance. In general, the spin system has magnetic sublattices $\sigma = a, b, c, \ldots$. The bosonic annihilation operator $\hat{a}_i$ is defined with two indices $i = (i, \sigma)$, where σ denotes the magnetic sublattice and i is the unit-cell index. For example, $\hat{b}_{(i,a)} = \hat{a}_i$ and $\hat{b}_{(i,b)} = \hat{b}_i$. This allows the operators to be transformed into their momentum-space representations:

$$\hat{b}_{i\sigma} = \frac{1}{\sqrt{L}} \sum_{\mathbf{k}\in \mathrm{FBZ}} e^{i\mathbf{k}\cdot\mathbf{r}_{i\sigma}} \hat{b}_{\mathbf{k}\sigma}, \tag{39}$$

where L is the number of unit cells in the system and FBZ denotes the first Brillouin zone. Then, the Hamiltonian becomes:

$$\begin{aligned} H_2 =& \sum_{\mathbf{k}} \Big[\sum_{\mathsf{L}(i\sigma,j\rho)} t_{ij}^{+-} e^{-i\mathbf{k}\cdot\boldsymbol{\delta}_{ij}} \hat{a}_{\mathbf{k}\sigma}^\dagger \hat{a}_{\mathbf{k}\rho} + t_{ij}^{-+} e^{i\mathbf{k}\cdot\boldsymbol{\delta}_{ij}} \hat{a}_{\mathbf{k}\sigma} \hat{a}_{\mathbf{k}\rho}^\dagger + t_{ij}^{++} e^{-i\mathbf{k}\cdot\boldsymbol{\delta}_{ij}} \hat{a}_{\mathbf{k}\sigma}^\dagger \hat{a}_{-\mathbf{k}\rho}^\dagger + t_{ij}^{--} e^{i\mathbf{k}\cdot\boldsymbol{\delta}_{ij}} \hat{a}_{\mathbf{k}\sigma} \hat{a}_{-\mathbf{k}\rho} \Big] + \sum_{\mathbf{k},\sigma} \mu_\sigma \hat{a}_{\mathbf{k}\sigma}^\dagger \hat{a}_{\mathbf{k}\sigma} \\ =& \frac{1}{2} \sum_{\mathbf{k}\in\mathrm{FBZ}} \begin{pmatrix} \psi_{\mathbf{k}}^\dagger & \psi_{-\mathbf{k}} \end{pmatrix} \begin{pmatrix} \mathsf{A}_{\mathbf{k}} & \mathsf{B}_{\mathbf{k}} \\ \mathsf{B}_{\mathbf{k}}^\dagger & \mathsf{A}_{-\mathbf{k}}^* \end{pmatrix} \begin{pmatrix} \psi_{\mathbf{k}} \\ \psi_{-\mathbf{k}}^\dagger \end{pmatrix} + \frac{1}{2} \sum_{\mathbf{k}\in\mathrm{FBZ}} C_{\mathbf{k}}, \end{aligned} \tag{40}$$

where $\psi_{\mathbf{k}}^\dagger = (a_{\mathbf{k}}^\dagger, b_{\mathbf{k}}^\dagger, \cdots)$ is a vector of bosonic creation operators for each sublattice, and the constant $C_{\mathbf{k}}$ follows from the normal ordering $\hat{a}_{\mathbf{k}\rho}^\dagger \hat{a}_{\mathbf{k}\sigma} = \frac{1}{2}(\hat{a}_{\mathbf{k}\rho}^\dagger \hat{a}_{\mathbf{k}\sigma} + \hat{a}_{\mathbf{k}\rho} \hat{a}_{\mathbf{k}\sigma}^\dagger) - \frac{1}{2}\delta_{\rho\sigma}$.

$$\begin{aligned} \sum_{\mathbf{k}\in\mathrm{FBZ}} C_{\mathbf{k}} =& \sum_{\mathbf{k}\in\mathrm{FBZ}} \Big[\sum_{\mathsf{L}(i\sigma,j\rho)} \big(t_{ij}^{-+} e^{i\mathbf{k}\cdot\boldsymbol{\delta}_{ij}} - t_{ij}^{+-} e^{-i\mathbf{k}\cdot\boldsymbol{\delta}_{ij}}\big)\delta_{\rho\sigma} - \sum_{\sigma} \mu_\sigma \Big] = -L \sum_{\sigma=a,b,\cdots} \mu_\sigma \\ =& L \sum_{\sigma=a,b,\cdots} \Big(\sum_{\rho\in\mathsf{L}(\sigma)} S_\rho \widetilde{J}_{\rho\sigma}^{00} - \widetilde{h}_\sigma^0 \Big) = \sum_i \frac{\partial E_{\rm cl}}{\partial \mathbf{S}_i} \cdot \hat{\mathbf{e}}_i^0 \end{aligned} \tag{41}$$

In the first line, the first term always vanishes if $\rho \neq \sigma$. Even when $\rho = \sigma$ for hopping within the same sublattice but between different unit cells, these terms vanish in the summation over the FBZ. **[REVIEW: A10: Claim that same-sublattice terms vanish under FBZ sum is non-trivial when $t^{-+} \neq (t^{+-})^*$ (e.g., DM interactions). Add justification or note conditions.]** Note that the constant term can be expressed as $\frac{1}{2}\sum_{\mathbf{k}\in\mathrm{FBZ}} C_{\mathbf{k}} = -\frac{1}{4}\mathrm{Tr}[\mathsf{H}_{\mathbf{k}}]$ **[REVIEW: A5: States $\frac{1}{2}\sum C_k = -\frac{1}{4}\mathrm{Tr}[H_k]$, but H_k includes A and B blocks. Since $\mathrm{Tr}[H_k] = 2\,\mathrm{Tr}[A_k]$ for Hermitian A, the factor should be $-\frac{1}{2}$ not $-\frac{1}{4}$. Verify.]** and

$$\frac{1}{2}\sum_{\mathbf{k}\in\mathrm{FBZ}} C_{\mathbf{k}} = \frac{1}{2}\sum_i \frac{\partial E_{\rm cl}}{\partial \mathbf{S}_i}\cdot\hat{\mathbf{e}}_i^0 = \frac{1}{2}\sum_i \Big(\widetilde{h}_i^0 - \sum_{j\in\mathsf{L}(i)} S_j \cdot \widetilde{J}_{ji}^{00}\Big) \xrightarrow{S_i=S} \frac{1}{S}\big(E_{\rm exc} + \frac{1}{2}E_{\rm zm}\big), \tag{42}$$

where $E_{\rm exc} = \sum_{ij} \mathbf{S}_i \cdot \mathbf{J}_{ij} \cdot \mathbf{S}_j$ is the exchange energy and $E_{\rm zm} = -\sum_i \mathbf{h}_i \cdot \mathbf{S}_i$ is the Zeeman energy. This is the reason why the quantum correction reduces the classical energy. Clearly, when $\mathbf{h} = 0$, we have

$$\hat{H} = S(S+1)E_{\rm cl} + SH_2 + S^{1/2}H_3 + \cdots. \tag{43}$$

[REVIEW: A11: $S(S+1)$ prefactor requires zero-point correction $= (1/S)E_{\rm exc}$. Derivation is omitted and non-trivial. Add sketch or reference.]

To summarize, the LSWT Hamiltonian can be expressed in the following form:

$$H^{(2)} = \frac{1}{2}\sum_{\mathbf{k}} \Big(\Psi_{\mathbf{k}}^\dagger \hat{\mathsf{H}}_{\mathbf{k}} \Psi_{\mathbf{k}} - \mathrm{Tr}[\mathsf{A}_{\mathbf{k}}] \Big) \tag{44}$$

where $\Psi_{\mathbf{k}}^\dagger = \Big(a_{\mathbf{k}}^\dagger, b_{\mathbf{k}}^\dagger, \ldots, a_{-\mathbf{k}}, b_{-\mathbf{k}}, \ldots\Big)$ is a vector of length $2M_s$, with M_s being the number of magnetic sublattices, and $\hat{\mathsf{H}}_{\mathbf{k}}$ is a $2M_s \times 2M_s$ matrix given by:

$$\hat{\mathsf{H}}_{\mathbf{k}} = \begin{pmatrix} \hat{\mathsf{A}}_{\mathbf{k}} & \hat{\mathsf{B}}_{\mathbf{k}} \\ \hat{\mathsf{B}}_{\mathbf{k}}^\dagger & \hat{\mathsf{A}}_{-\mathbf{k}}^* \end{pmatrix}, \tag{45}$$

where $\hat{\mathsf{A}}^\dagger_{\mathbf{k}} = \hat{\mathsf{A}}_{\mathbf{k}}$ are Hermitian, which follows from the Hermiticity of the Hamiltonian, and $\hat{\mathsf{B}}_{-\mathbf{k}} = \hat{\mathsf{B}}^{\mathrm{T}}_{\mathbf{k}}$, which follows from the bosonic commutation relations.

Finally, we provide explicit expressions for $\mathsf{H}_{\mathbf{k}}$ for a two-sublattice system $(\rho,\sigma) = (a,b)$ for the reader's convenience. Let us first consider the case where $a \neq b$, which can be written in matrix form as:

(SJ: I think t^{--}_{ij} should be replaced by $(t^{--}_{ij})^*$ in equation 46, to satisfy the constraint on B, $\hat{\mathsf{B}}_{-\mathbf{k}} = \hat{\mathsf{B}}^{\mathrm{T}}_{\mathbf{k}}$)

(SP): I agree and this is a typo. The code is implemented without this typo. **[REVIEW: A1: SJ/SP comments should be removed for final version. The typo (t^{++} in both off-diagonals of B_k instead of one being $(t^{--})^*$) is confirmed; fix the equation accordingly.] [REVIEW: B10: Remove all \textcolor{blue/red}{(SJ/SP:...)} annotations before final version.]**

$$\mathsf{A}_{\mathbf{k}} = \begin{pmatrix} \widetilde{h}_a - \sum_{\mathsf{L}(i,j)} \widetilde{J}^{00}_{ij} S_b & \sum_{\mathsf{L}(i,j)} t^{+-}_{ij} e^{-i\mathbf{k}\cdot\boldsymbol{\delta}_{ij}} \\ \sum_{\mathsf{L}(i,j)} t^{-+}_{ij} e^{i\mathbf{k}\cdot\boldsymbol{\delta}_{ij}} & \widetilde{h}_b - \sum_{\mathsf{L}(i,j)} \widetilde{J}^{00}_{ij} S_a \end{pmatrix}, \quad \mathsf{B}_{\mathbf{k}} = \begin{pmatrix} 0 & \sum_{\mathsf{L}(i,j)} t^{++}_{ij} e^{-i\mathbf{k}\cdot\boldsymbol{\delta}_{ij}} \\ \sum_{\mathsf{L}(i,j)} t^{++}_{ij} e^{i\mathbf{k}\cdot\boldsymbol{\delta}_{ij}} & 0 \end{pmatrix}, \tag{46}$$

If the two sublattices are identical ($a = b$), the matrices in the Hamiltonian are given as: (SJ: I think the factor 2 in A_k (in equation 47) is unnecessary; the chemical potential is doublely counted for A_k and A_{-k}, and divided by 2 in total LSWT Hamiltonian) **[REVIEW: A2: SJ comment on factor 2: since LSWT Hamiltonian is $(1/2)\sum \Psi^\dagger H \Psi$, both A_k and A^*_{-k} contribute chemical potential once each. The explicit factor 2 in A_k may cause double counting. Verify and correct.] [REVIEW: B10: Remove SJ colored annotation before final version.]**

$$\begin{aligned} \mathsf{A}_{\mathbf{k}} =& 2\Big(\widetilde{h}_a - \sum_{\mathsf{L}(ij)} \widetilde{J}^{00}_{ij} S_a\Big) + \sum_{\mathsf{L}(i,j)} (t^{+-}_{ij} e^{-i\mathbf{k}\cdot\boldsymbol{\delta}_{ij}} + t^{-+}_{ij} e^{+i\mathbf{k}\cdot\boldsymbol{\delta}_{ij}}), \\ \mathsf{B}_{\mathbf{k}} =& \sum_{\mathsf{L}(i,j)} t^{++}_{ij} (e^{-i\mathbf{k}\cdot\boldsymbol{\delta}_{ij}} + e^{+i\mathbf{k}\cdot\boldsymbol{\delta}_{ij}}) = 2 \sum_{\mathsf{L}(i,j)} t^{++}_{ij} \cos(\mathbf{k}\cdot\boldsymbol{\delta}_{ij}) \end{aligned} \tag{47}$$

This momentum-space formulation simplifies the analysis of bosonic excitations and provides a basis for studying dispersion relations and system stability.

E. Diagonalization of Quadratic Boson Hamiltonian

In this section, we discuss the diagonalization of the boson Hamiltonian. The quadratic boson Hamiltonian can be diagonalized by the Bogoliubov transformation. Consider the following Bogoliubov transformation:

$$\hat{b}_{\mathbf{k},\mu} = \big[\mathsf{P}_{\mathbf{k}} \hat{\boldsymbol{\beta}}_{\mathbf{k}} + \mathsf{Q}_{-\mathbf{k}} \hat{\boldsymbol{\beta}}^\dagger_{-\mathbf{k}}\big]_\mu = \sum_{\nu=1}^{M_s} [\mathsf{P}_{\mathbf{k}}]_{\mu\nu} \hat{\beta}_{\mathbf{k},\nu} + [\mathsf{Q}_{-\mathbf{k}}]_{\mu\nu} \hat{\beta}^\dagger_{-\mathbf{k},\nu}, \tag{48}$$

where the indices μ,ν are defined over $\mu,\nu = 1,\cdots,M_s$. In order to preserve the bosonic commutation relations, we require the following identities:

$$\begin{aligned} [\hat{b}_{\mathbf{k},\mu}, \hat{b}^\dagger_{\mathbf{q},\nu}] =& \big[\mathsf{P}_{\mathbf{k}}\mathsf{P}^\dagger_{\mathbf{k}} - \mathsf{Q}_{-\mathbf{k}}\mathsf{Q}^\dagger_{-\mathbf{k}}\big]_{\mu\nu} \delta_{\mathbf{k},\mathbf{q}} = +\delta_{\mu,\nu}\delta_{\mathbf{k},\mathbf{q}}, \qquad & [\hat{b}_{\mathbf{k},\mu}, \hat{b}_{-\mathbf{q},\nu}] = \big[\mathsf{P}_{\mathbf{k}}\mathsf{Q}^{\mathrm{T}}_{\mathbf{k}} - \mathsf{Q}_{-\mathbf{k}}\mathsf{P}^{\mathrm{T}}_{-\mathbf{k}}\big]_{\mu\nu} \delta_{\mathbf{k},\mathbf{q}} = 0, \\ [\hat{b}^\dagger_{-\mathbf{k},\mu}, \hat{b}_{-\mathbf{q},\nu}] =& \big[\mathsf{P}^*_{-\mathbf{k}}\mathsf{P}^{\mathrm{T}}_{-\mathbf{k}} - \mathsf{Q}^*_{\mathbf{k}}\mathsf{Q}^{\mathrm{T}}_{\mathbf{k}}\big]_{\mu\nu} \delta_{\mathbf{k},\mathbf{q}} = -\delta_{\mu,\nu}\delta_{\mathbf{k},\mathbf{q}}, \qquad & [\hat{b}^\dagger_{-\mathbf{k},\mu}, \hat{b}^\dagger_{\mathbf{q},\nu}] = \big[\mathsf{Q}^*_{\mathbf{k}}\mathsf{P}^\dagger_{\mathbf{k}} - \mathsf{P}^*_{-\mathbf{k}}\mathsf{Q}^\dagger_{-\mathbf{k}}\big]_{\mu\nu} \delta_{\mathbf{k},\mathbf{q}} = 0. \end{aligned}$$

These identities can be represented in a compact matrix form:

$$\begin{pmatrix} \mathsf{P}_{\mathbf{k}} & \mathsf{Q}_{-\mathbf{k}} \\ \mathsf{Q}^*_{\mathbf{k}} & \mathsf{P}^*_{-\mathbf{k}} \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} \mathsf{P}^\dagger_{\mathbf{k}} & \mathsf{Q}^{\mathrm{T}}_{\mathbf{k}} \\ \mathsf{Q}^\dagger_{-\mathbf{k}} & \mathsf{P}^{\mathrm{T}}_{-\mathbf{k}} \end{pmatrix} = \begin{pmatrix} \mathsf{P}_{\mathbf{k}}\mathsf{P}^\dagger_{\mathbf{k}} - \mathsf{Q}_{-\mathbf{k}}\mathsf{Q}^\dagger_{-\mathbf{k}} & \mathsf{P}_{\mathbf{k}}\mathsf{Q}^{\mathrm{T}}_{\mathbf{k}} - \mathsf{Q}_{-\mathbf{k}}\mathsf{P}^{\mathrm{T}}_{-\mathbf{k}} \\ \mathsf{Q}^*_{\mathbf{k}}\mathsf{P}^\dagger_{\mathbf{k}} - \mathsf{P}^*_{-\mathbf{k}}\mathsf{Q}^\dagger_{-\mathbf{k}} & \mathsf{Q}^*_{\mathbf{k}}\mathsf{Q}^{\mathrm{T}}_{\mathbf{k}} - \mathsf{P}^*_{-\mathbf{k}}\mathsf{P}^{\mathrm{T}}_{-\mathbf{k}} \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \tag{49}$$

By adopting the above convention (i.e., the transformation of $\hat{\boldsymbol{b}}_{\mathbf{k}} \to \hat{\boldsymbol{b}}^\dagger_{-\mathbf{k}}$), the (para)-unitary transformation matrix $\mathsf{T}_{\mathbf{k}}$ has the following form:

$$\Psi_{\mathbf{k}} = \mathsf{T}_{\mathbf{k}} \widetilde{\Psi}_{\mathbf{k}} : \quad \begin{pmatrix} \hat{\boldsymbol{b}}_{\mathbf{k}} \\ \hat{\boldsymbol{b}}^\dagger_{-\mathbf{k}} \end{pmatrix} = \begin{pmatrix} \mathsf{P}_{\mathbf{k}} & \mathsf{Q}_{-\mathbf{k}} \\ \mathsf{Q}^*_{\mathbf{k}} & \mathsf{P}^*_{-\mathbf{k}} \end{pmatrix} \begin{pmatrix} \hat{\boldsymbol{\beta}}_{\mathbf{k}} \\ \hat{\boldsymbol{\beta}}^\dagger_{-\mathbf{k}} \end{pmatrix}. \tag{50}$$

Suppose the unitary transformation $\mathsf{T}_{\mathbf{k}}$ diagonalizes the Hamiltonian matrix $\mathsf{H}_{\mathbf{k}}$:

$$\mathsf{T}^\dagger_{\mathbf{k}} \mathsf{H}_{\mathbf{k}} \mathsf{T}_{\mathbf{k}} = \mathsf{E}_{\mathbf{k}} : \qquad \begin{pmatrix} \mathsf{P}^\dagger_{\mathbf{k}} & \mathsf{Q}^{\mathrm{T}}_{\mathbf{k}} \\ \mathsf{Q}^\dagger_{-\mathbf{k}} & \mathsf{P}^{\mathrm{T}}_{-\mathbf{k}} \end{pmatrix} \begin{pmatrix} \mathsf{A}_{\mathbf{k}} & \mathsf{B}_{\mathbf{k}} \\ \mathsf{B}^\dagger_{\mathbf{k}} & \mathsf{A}^*_{-\mathbf{k}} \end{pmatrix} \begin{pmatrix} \mathsf{P}_{\mathbf{k}} & \mathsf{Q}_{-\mathbf{k}} \\ \mathsf{Q}^*_{\mathbf{k}} & \mathsf{P}^*_{-\mathbf{k}} \end{pmatrix} = \begin{pmatrix} \mathsf{E}_{\mathbf{k}} & 0 \\ 0 & \widetilde{\mathsf{E}}_{-\mathbf{k}} \end{pmatrix}, \tag{51}$$

The two diagonal matrices $\mathsf{E}_{\mathbf{k}}$ and $\widetilde{\mathsf{E}}_{-\mathbf{k}}$ are real-valued, and their explicit forms are written as:

$$\begin{aligned}\mathsf{E}_{\mathbf{k}} =& \mathsf{P}^{\dagger}_{\mathbf{k}}\mathsf{A}_{\mathbf{k}}\mathsf{P}_{\mathbf{k}} + \mathsf{P}^{\dagger}_{\mathbf{k}}\mathsf{B}_{\mathbf{k}}\mathsf{Q}^{*}_{\mathbf{k}} + \mathsf{Q}^{\mathrm{T}}_{\mathbf{k}}\mathsf{B}^{\dagger}_{\mathbf{k}}\mathsf{P}_{\mathbf{k}} + \mathsf{Q}^{\mathrm{T}}_{\mathbf{k}}\mathsf{A}^{*}_{-\mathbf{k}}\mathsf{Q}^{*}_{\mathbf{k}},\\ \widetilde{\mathsf{E}}_{-\mathbf{k}} =& \mathsf{Q}^{\dagger}_{-\mathbf{k}}\mathsf{A}_{\mathbf{k}}\mathsf{Q}_{-\mathbf{k}} + \mathsf{Q}^{\dagger}_{-\mathbf{k}}\mathsf{B}_{\mathbf{k}}\mathsf{P}^{*}_{-\mathbf{k}} + \mathsf{P}^{\mathrm{T}}_{-\mathbf{k}}\mathsf{B}^{\dagger}_{\mathbf{k}}\mathsf{Q}_{-\mathbf{k}} + \mathsf{P}^{\mathrm{T}}_{-\mathbf{k}}\mathsf{A}^{*}_{-\mathbf{k}}\mathsf{P}^{*}_{-\mathbf{k}}.\end{aligned}$$

Note that the two diagonal matrices can be mapped to each other under conjugation and momentum inversion. Thus, we obtain the energy bands ($E_{\mathbf{k}}$ and $E_{-\mathbf{k}}$) when we diagonalize the quadratic boson Hamiltonian via Colpa's method [**?**]:

$$\widetilde{\mathsf{E}}_{-\mathbf{k}} \xrightarrow[\mathbf{k}\to-\mathbf{k}]{\text{conj}} \widetilde{\mathsf{E}}^{*}_{\mathbf{k}}: \qquad \widetilde{\mathbf{E}}^{*}_{\mathbf{k}} = \mathsf{P}^{\dagger}_{\mathbf{k}}\mathsf{A}_{\mathbf{k}}\mathsf{P}_{\mathbf{k}} + \mathsf{P}^{\dagger}_{\mathbf{k}}\mathsf{B}^{\mathrm{T}}_{-\mathbf{k}}\mathsf{Q}^{*}_{\mathbf{k}} + \mathsf{Q}^{\mathrm{T}}_{\mathbf{k}}\mathsf{B}^{*}_{-\mathbf{k}}\mathsf{P}_{\mathbf{k}} + \mathsf{Q}^{\mathrm{T}}_{\mathbf{k}}\mathsf{A}^{*}_{-\mathbf{k}}\mathsf{Q}^{*}_{\mathbf{k}} = \mathbf{E}_{\mathbf{k}}, \tag{52}$$

where $\mathsf{B}^{\mathrm{T}}_{-\mathbf{k}} = \mathsf{B}_{\mathbf{k}}$ and $\mathsf{B}^{*}_{-\mathbf{k}} = \mathsf{B}^{\dagger}_{\mathbf{k}}$. Since $\mathsf{E}_{\mathbf{k}}$ is a *real* diagonal matrix, we conclude $\widetilde{\mathsf{E}}_{\mathbf{k}} = \mathsf{E}_{\mathbf{k}}$. Thus, the canonical form of the quadratic bosonic Hamiltonian reads:

$$\begin{aligned}\hat{H}_2 =& \frac{1}{2}\sum_{\mathbf{k}}\Big((\widetilde{\Psi}^{\dagger}_{\mathbf{k}}\mathsf{T}^{\dagger}_{\mathbf{k}})\mathsf{H}_{\mathbf{k}}(\mathsf{T}_{\mathbf{k}}\widetilde{\Psi}_{\mathbf{k}}) - \mathrm{Tr}[\mathsf{A}_{\mathbf{k}}]\Big) = \frac{1}{2}\sum_{\mathbf{k}}\Big(\widetilde{\Psi}^{\dagger}_{\mathbf{k}}\mathsf{E}_{\mathbf{k}}\widetilde{\Psi}_{\mathbf{k}} - \mathrm{Tr}[\mathsf{A}_{\mathbf{k}}]\Big)\\ =& \frac{1}{2}\sum_{\mathbf{k}}\begin{pmatrix}\hat{\boldsymbol{\beta}}_{\mathbf{k}}\\ \hat{\boldsymbol{\beta}}^{\dagger}_{-\mathbf{k}}\end{pmatrix}^{\dagger}\begin{pmatrix}\mathbf{E}_{\mathbf{k}} & 0\\ 0 & \widetilde{\mathbf{E}}_{-\mathbf{k}}\end{pmatrix}\begin{pmatrix}\hat{\boldsymbol{\beta}}_{\mathbf{k}}\\ \hat{\boldsymbol{\beta}}^{\dagger}_{-\mathbf{k}}\end{pmatrix} - \frac{1}{2}\mathrm{Tr}[\mathsf{A}_{\mathbf{k}}]\\ =& \frac{1}{2}\sum_{\mathbf{k},\sigma}\left(E_{\mathbf{k},\sigma}\hat{\beta}^{\dagger}_{\mathbf{k},\sigma}\hat{\beta}_{\mathbf{k},\sigma} + E_{-\mathbf{k},\sigma}\hat{\beta}_{-\mathbf{k},\sigma}\hat{\beta}^{\dagger}_{-\mathbf{k},\sigma}\right) - \frac{1}{2}\mathrm{Tr}[\mathsf{A}_{\mathbf{k}}] \qquad (53)\\ =& \sum_{\mathbf{k},\sigma}E_{\mathbf{k},\sigma}\hat{\beta}^{\dagger}_{\mathbf{k},\sigma}\hat{\beta}_{\mathbf{k},\sigma} + \frac{1}{2}\sum_{\mathbf{k},\sigma}\left(E_{\mathbf{k},\sigma} - \mathrm{Tr}[\mathsf{A}_{\mathbf{k}}]\right), \qquad (54)\end{aligned}$$

where $E_{\mathbf{k}} \geq 0$ for all $\mathbf{k} \in \mathrm{BZ}$. The constant term is called the "zero-point energy," as it represents the ground state energy of the quantum system at zero temperature (the Gibbs state at $T = 0$). **[REVIEW: A12: E_0 includes E_{cl}. Conventionally 'zero-point energy' refers only to $\frac{1}{2}\sum E_k$. Consider 'ground state energy' for E_0 and 'zero-point correction' for the quantum part.]** This gives the quantum correction to the classical energy in the LSWT framework:

$$E_0 = E_{\mathrm{cl}} + \frac{1}{2}\sum_{\mathbf{k}\in\mathrm{FBZ}}\Big(\sum_{\sigma=a,b,\cdots}E_{\mathbf{k},\sigma} - \mathrm{Tr}[\mathsf{A}_{\mathbf{k}}]\Big), \tag{55}$$

and the energy density per spin site is:

$$\mathcal{E} = \frac{E_0}{N} = \frac{1}{N}\Big[E_{\mathrm{cl}} + \frac{1}{2}\sum_{\mathbf{k}\in\mathrm{FBZ}}\Big(\sum_{\sigma=a,b,\cdots}E_{\mathbf{k},\sigma} - \mathrm{Tr}[\mathsf{A}_{\mathbf{k}}]\Big)\Big], \tag{56}$$

where $N = Lm_s$ is the total number of spin sites in the system, with L being the number of unit cells and m_s the number of magnetic sublattices per unit cell.

F. Paraunitary Diagonalization

Now, the remaining problem is to find a paraunitary matrix $\mathsf{T}_{\mathbf{k}}$ (we will suppress the $\mathbf{k}$ subscript in this section for clarity, so $\mathsf{H}_{\mathbf{k}} \to \mathsf{H}$, etc.) such that

$$\mathsf{T}^{\dagger}\mathsf{H}\mathsf{T} = \mathsf{E}, \quad \text{with} \quad \mathsf{T}\mathsf{J}\mathsf{T}^{\dagger} = \mathsf{J}, \tag{57}$$

where E is a diagonal matrix containing the magnon energies, and J is the symplectic metric tensor. The paraunitary transformation T exists if and only if H is positive definite. **[REVIEW: B18: Colpa requires H positive definite. For positive semi-definite (Goldstone modes), modified procedures exist. Add caveat.]**

The explicit form of the paraunitary matrix is given by

$$\mathsf{T} = (\mathsf{K}^{\dagger})^{-1}\mathsf{V}(\Lambda\mathsf{J})^{1/2}, \tag{58}$$

where K comes from the Cholesky decomposition of the Hamiltonian. Since the Hamiltonian matrix H is positive definite, the Cholesky decomposition exists, satisfying:

$$\mathsf{H} = \mathsf{K}\mathsf{K}^\dagger, \tag{59}$$

and V is the unitary matrix that diagonalizes $\mathsf{K}^\dagger \mathsf{J}\mathsf{K}$:

$$\mathsf{V}\Lambda\mathsf{V}^\dagger = \mathsf{K}^\dagger\mathsf{J}\mathsf{K}, \tag{60}$$

where Λ is a diagonal matrix.

We will now verify that this construction satisfies the required paraunitary conditions. First, let us check $\mathsf{T}\mathsf{J}\mathsf{T}^\dagger = \mathsf{J}$:

$$\begin{aligned}
\mathsf{T}\mathsf{J}\mathsf{T}^\dagger &= \big((\mathsf{K}^\dagger)^{-1}\mathsf{V}(\Lambda\mathsf{J})^{1/2}\big)\mathsf{J}\big((\Lambda\mathsf{J})^{1/2}\big)^\dagger\mathsf{V}^\dagger\mathsf{K}^{-1} \\
&= (\mathsf{K}^\dagger)^{-1}\mathsf{V}(\Lambda\mathsf{J})^{1/2}\mathsf{J}(\mathsf{J}\Lambda)^{1/2}\mathsf{V}^\dagger\mathsf{K}^{-1} \\
&= (\mathsf{K}^\dagger)^{-1}\mathsf{V}\Lambda\mathsf{V}^\dagger\mathsf{K}^{-1} \\
&= (\mathsf{K}^\dagger)^{-1}(\mathsf{K}^\dagger\mathsf{J}\mathsf{K})\mathsf{K}^{-1} = \mathsf{J}
\end{aligned}$$

where we can also verify that $\mathsf{T}^\dagger\mathsf{J}\mathsf{T} = \mathsf{J}$.

Next, let us check that T diagonalizes the Hamiltonian as required:

$$\begin{aligned}
\mathsf{T}^\dagger\mathsf{H}\mathsf{T} &= \big((\Lambda\mathsf{J})^{1/2}\big)^\dagger\mathsf{V}^\dagger\mathsf{K}^{-1}\cdot\mathsf{K}\mathsf{K}^\dagger\cdot(\mathsf{K}^\dagger)^{-1}\mathsf{V}(\Lambda\mathsf{J})^{1/2} \\
&= (\mathsf{J}\Lambda)^{1/2}\mathsf{V}^\dagger\mathsf{V}(\Lambda\mathsf{J})^{1/2} \\
&= (\mathsf{J}\Lambda)^{1/2}(\Lambda\mathsf{J})^{1/2} \\
&= \mathsf{J}\Lambda = \Lambda\mathsf{J} = \mathsf{E}
\end{aligned}$$

where we identify the diagonal energy matrix as $\mathsf{E} = \mathsf{J}\Lambda = \Lambda\mathsf{J}$. This equality holds because both J and Λ are diagonal matrices, with J having half positive and half negative entries, and Λ arranged to ensure that E is always positive definite.

II. PHYSICAL QUANTITIES IN LINEAR SPIN WAVE THEORY

In this Section, I summarize the methods for obtaining important physical quantities in the frame of linear spin wave theory. **[REVIEW: B11: 'I summarize' → 'we summarize'. Unify first-person convention throughout.]** The physical quantities computable in LSWT are the following:

Quantity	Definition	Calculation in LSWT	Sect.
		Thermodynamics	**III**
Z_β	$\mathrm{Tr}[e^{-\beta\hat{H}}]$	$e^{-\beta E_0}\prod_{\mathbf{k},\mu}\frac{1}{1-e^{-\beta E_{\mathbf{k},\mu}}}$	III A
U	$\langle\hat{H}\rangle$	$\sum_{\mathbf{k},\mu} E_{\mathbf{k},\mu} n_{\mathbf{k},\mu}$	III B
F	$-k_B T\ln Z_\beta$	$k_B T\sum_{\mathbf{k},\mu}\ln\left(1-e^{-\beta E_{\mathbf{k},\mu}}\right)+E_0$	III C
S	$-k_{\mathrm{B}}\mathrm{Tr}\big(\rho\ln\rho\big)$	$k_B\sum_{\mathbf{k},\mu}[(1+n_{\mathbf{k},\mu})\ln(1+n_{\mathbf{k},\mu})-n_{\mathbf{k},\mu}\ln n_{\mathbf{k},\mu}]$	III D
C	$\frac{\partial U}{\partial T}$	$\sum_{\mathbf{k},\mu}\frac{\partial n_{\mathbf{k},\mu}}{\partial T}E_{\mathbf{k},\mu}$	III E
$\langle\hat{n}_\mu\rangle$	$\sum_{\mathbf{k}}\langle a^\dagger_{\mathbf{k},\mu}a_{\mathbf{k},\mu}\rangle$	$\sum_{\mathbf{k}}[\langle\Psi_{\mathbf{k}}\Psi^\dagger_{\mathbf{k}}\rangle]_{m_s+\mu,m_s+\mu}$	III F
$\langle\widetilde{S}_\mu\rangle$	$S_\mu-\langle\hat{n}_\mu\rangle$	$S_\mu-\frac{1}{L}\sum_{\mathbf{k}}\langle\hat{n}_{\mathbf{k},\mu}\rangle$	III F
		Correlations	**IV**
$\mathcal{C}^{\alpha\beta}_{\mu\nu}(\mathbf{k},t)$	$\langle S^\alpha_{\mathbf{k},\mu}(t)S^\beta_{-\mathbf{k},\nu}(0)\rangle$	$\sum_{(m,n)_{\mu\nu}}[\mathsf{U}^\alpha\mathsf{S}_{\mathbf{k}}\mathsf{T}_{\mathbf{k}}\mathsf{N}_{\mathbf{k}}(t)\mathsf{T}^\dagger_{\mathbf{k}}\bar{\mathsf{S}}_{\mathbf{k}}\bar{\mathsf{U}}^\beta]_{mn}$	IV A
$\mathcal{S}^{\alpha\beta}(\mathbf{k},\omega)$	$\frac{1}{2\pi}\int dt\, e^{i\omega t}\mathcal{C}^{\alpha\beta}(\mathbf{k},t)$	$\sum_{(m,n)_{\mu\nu}}[\mathsf{U}^\alpha\mathsf{S}_{\mathbf{k}}\mathsf{T}_{\mathbf{k}}\mathcal{F}[\mathsf{N}_{\mathbf{k}}](\omega;\eta)\mathsf{T}^\dagger_{\mathbf{k}}\bar{\mathsf{S}}_{\mathbf{k}}\bar{\mathsf{U}}^\beta]_{m,n}$	IV B
$\mathcal{A}^{\alpha\beta}(\mathbf{k},\omega)$	$-\frac{1}{\pi}\,\mathrm{Im}[G^{\alpha\beta}_{\mathrm{R}}(\mathbf{k},\omega)]$	$-\frac{2}{\pi}\,\mathrm{Im}\sum_{m,n}[\mathsf{U}^\alpha\mathsf{S}_{\mathbf{k}}\mathsf{T}_{\mathbf{k}}\mathcal{G}[\mathsf{N}_{\mathbf{k}}](\omega;\eta)\mathsf{T}^\dagger_{\mathbf{k}}\mathsf{S}_{\mathbf{k}}\mathsf{U}^\beta]_{m,n}$	IV C
		Topology	**V**
Q_{skyr}	$\frac{1}{4\pi}\sum_\triangle\chi_\triangle$	$\chi_\triangle=2\arctan\left(\frac{\lvert\mathbf{m}_i\cdot(\mathbf{m}_j\times\mathbf{m}_k)\rvert}{1+\mathbf{m}_i\cdot\mathbf{m}_j+\mathbf{m}_j\cdot\mathbf{m}_k+\mathbf{m}_k\cdot\mathbf{m}_i}\right)$	V A
C_n	$\frac{1}{2\pi}\int_{\mathrm{FBZ}}\Omega_{n,\mathbf{k}}\,d^2\mathbf{k}$	$\Omega_{n,\mathbf{k}}=-2\,\mathrm{Im}\sum_{m\neq n}\frac{[\mathsf{J}\mathsf{T}^\dagger_{\mathbf{k}}(\partial_x\mathsf{H}_{\mathbf{k}})\mathsf{T}_{\mathbf{k}}]_{nm}[\mathsf{J}\mathsf{T}^\dagger_{\mathbf{k}}(\partial_y\mathsf{H}_{\mathbf{k}})\mathsf{T}_{\mathbf{k}}]_{mn}}{(E_{\mathbf{k},nn}-E_{\mathbf{k},mm})^2}$	V B
κ_{xy}	Thermal Hall conductivity	$-\frac{k_B^2T}{V}\sum_n\int_{\mathbf{k}}d^2\mathbf{k}\left(c_2(n_{\mathbf{k},n})-\frac{\pi^2}{3}\right)\Omega_{n,\mathbf{k}}$	V C

TABLE II. Physical Quantities in Linear Spin Wave Theory (LSWT). Here, $\beta=1/k_BT$, $n_{\mathbf{k},\mu}=\frac{1}{e^{\beta E_{\mathbf{k},\mu}}-1}$ is the Bose-Einstein distribution function, $\Omega_{n,\mathbf{k}}$ is the Berry curvature, and $c_2(x)$ is the Spence function.

[REVIEW: C14: Spectral function row uses S_k, U^β (no bar) while correlation row uses $\bar{S}_k$, $\bar{U}^\beta$, and body text uses $[R^\beta_k]^\dagger$. Unify notation.]

III. THERMODYNAMICS IN LINEAR SPIN WAVE THEORY

A. Partition function

In this section, we derive the partition function for linear spin wave theory. Recall that the Hamiltonian for the spin system with normal bosons is written as

$$\hat{H} = E_{\mathrm{cl}} + \sum_{\mathbf{k},\mu} E_{\mathbf{k},\mu}\Big(\beta^\dagger_{\mathbf{k},\mu}\beta_{\mathbf{k},\mu} + \frac{1}{2}\Big) - \frac{1}{2}\mathrm{Tr}[\mathsf{H}_{\mathbf{k}}] = E_0 + \sum_{\mathbf{k}\in\mathrm{FBZ}}\sum_{\mu=a,b,\cdots} E_{\mathbf{k},\mu}\hat{n}_{\mathbf{k},\mu}, \tag{61}$$

where $E_0 = E_{\mathrm{cl}} + \frac{1}{2}\sum_{\mathbf{k}}\big(\sum_\mu E_{\mathbf{k},\mu} - \mathrm{Tr}[\mathsf{A}_{\mathbf{k}}]\big)$ represents the zero-point energy [Eqn. (55)] and $\hat{n}_\mu(\mathbf{k}) = \beta^\dagger_{\mathbf{k},\mu}\beta_{\mathbf{k},\mu}$ is the magnon number operator. Here, $E_{\mathbf{k},\mu}$ are the magnon energy eigenvalues with band index μ.

The partition function is given by

$$Z_\beta \equiv \mathrm{Tr}\big(e^{-\beta\hat{H}}\big) = e^{-\beta E_0}\mathrm{Tr}\big(e^{-\beta\sum_{\mathbf{k},\mu}E_{\mathbf{k},\mu}\beta^\dagger_{\mathbf{k},\mu}\beta_{\mathbf{k},\mu}}\big) = e^{-\beta E_0}\mathrm{Tr}\big(e^{-\beta\sum_{\mathbf{k},\mu}E_{\mathbf{k},\mu}\hat{n}_{\mathbf{k},\mu}}\big), \tag{62}$$

Here, the constant term E_0 contributes to the internal energy but is not important for thermodynamic quantities that involve derivatives of $\ln\tilde{Z}_\beta$, where $\tilde{Z}_\beta$ represents the partition function without the zero-point energy contribution.

The partition function Z_β can be evaluated as

$$\mathrm{Tr}\big(e^{-\beta\sum_{\mathbf{k},\mu}E_{\mathbf{k},\mu}\hat{n}_\mu(\mathbf{k})}\big) = \mathrm{Tr}\Big[\prod_{\mathbf{k},\mu} e^{-\beta E_{\mathbf{k},\mu}\hat{n}_{\mathbf{k},\mu}}\Big] = \prod_{\mathbf{k},\mu}\mathrm{Tr}\Big[e^{-\beta E_{\mathbf{k},\mu}\hat{n}_{\mathbf{k},\mu}}\Big] = \prod_{\mathbf{k},\mu}\frac{1}{1-e^{-\beta E_{\mathbf{k},\mu}}} \tag{63a}$$

$$-\ln Z_\beta = \beta E_0 + \sum_{\mathbf{k},\mu}\ln\big(1-e^{-\beta E_{\mathbf{k},\mu}}\big). \tag{63b}$$

The final expression for $-\ln Z_\beta$ is particularly useful for calculating thermodynamic quantities such as free energy, internal energy, specific heat, and entropy in the linear spin-wave approximation.

B. Internal Energy

The internal energy U is the expectation value of the Hamiltonian $\hat{H}$, computed as:

$$U \equiv \langle\hat{H}\rangle = -\frac{\partial\ln Z_\beta}{\partial\beta}, \tag{64}$$

where $Z_\beta = e^{-\beta E_0}\tilde{Z}_\beta$, with E_0 being the zero-point energy and $\tilde{Z}_\beta$ the partition function excluding the zero-point energy contribution.

Computing the derivative, we get:

$$-\frac{\partial\ln Z_\beta}{\partial\beta} = E_0 + \frac{\partial}{\partial\beta}\Big[\sum_{\mathbf{k},\mu}\ln\big(1-e^{-\beta E_{\mathbf{k},\mu}}\big)\Big] = E_0 + \sum_{\mathbf{k},\mu}\frac{E_{\mathbf{k},\mu}e^{-\beta E_{\mathbf{k},\mu}}}{1-e^{-\beta E_{\mathbf{k},\mu}}} = E_0 + \sum_{\mathbf{k},\mu}E_{\mathbf{k},\mu}\frac{1}{e^{\beta E_{\mathbf{k},\mu}}-1}, \tag{65}$$

where we identify the Bose-Einstein distribution function:

$$n_\mu(\mathbf{k}) = \frac{1}{e^{\beta E_{\mathbf{k},\mu}}-1}, \tag{66}$$

which represents the thermal expectation value of the magnon number operator $\langle\hat{n}_{\mathbf{k},\mu}\rangle = \langle\hat{\beta}^\dagger_{\mathbf{k},\mu}\hat{\beta}_{\mathbf{k},\mu}\rangle$. Therefore, the internal energy can be expressed as:

$$\begin{aligned} U &= E_0 + \sum_{\mathbf{k},\mu}E_{\mathbf{k},\mu}n_\mu(\mathbf{k}) \\ &= E_{\mathrm{cl}} + \frac{1}{2}\sum_{\mathbf{k}}\big(\sum_\mu E_{\mathbf{k},\mu} - \mathrm{Tr}[\mathsf{A}_{\mathbf{k}}]\big) + \sum_{\mathbf{k},\mu}E_{\mathbf{k},\mu}n_\mu(\mathbf{k}) \end{aligned} \tag{67}$$

This result expresses the internal energy as the sum of the classical ground state energy, the zero-point quantum fluctuations, and the thermal contribution from magnon excitations.

C. Free Energy

The Helmholtz free energy (F) is defined as

$$Z_\beta \equiv e^{-\beta F} \tag{68}$$

where $Z_\beta = e^{-\beta E_0}\tilde{Z}_\beta$ is the partition function, with $\tilde{Z}_\beta$ being the partition function excluding the zero-point energy contribution. From (63) and Eq. (68), we get:

$$\begin{aligned}
F &= -\frac{1}{\beta}\ln(e^{-\beta E_0}\tilde{Z}_\beta) \\
&= E_0 - \frac{1}{\beta}\ln \tilde{Z}_\beta \\
&= E_{\rm cl} + \frac{1}{2}\sum_{\mathbf{k}}\big(\sum_{\mu} E_{\mathbf{k},\mu} - \mathrm{Tr}[\mathsf{A}_{\mathbf{k}}]\big) + \frac{1}{\beta}\sum_{\mathbf{k},\mu}\ln\left(1 - e^{-\beta E_{\mathbf{k},\mu}}\right),
\end{aligned} \tag{69}$$

where we use $-\ln \tilde{Z}_\beta = \sum_{\mathbf{k},\mu}\ln\left(1 - e^{-\beta E_{\mathbf{k},\mu}}\right)$ from the partition function section, and $E_0 = E_{\rm cl} + \frac{1}{2}\sum_{\mathbf{k}}\big(\sum_\mu E_{\mathbf{k},\mu} - \mathrm{Tr}[\mathsf{A}_{\mathbf{k}}]\big)$ is the zero-point energy.

D. Entropy Expression

In this section, we derive the expression for the thermal entropy in LSWT. The entropy is defined as

$$S \equiv -k_{\rm B}\mathrm{Tr}\big(\rho \ln \rho\big), \tag{70}$$

where ρ is the density operator, which is Gibbs states ($\rho = e^{-\beta \hat{H}}/Z_\beta$) especially in thermal equilibrium. **[REVIEW: B19: Rephrase tautological statement. Suggest: 'For Gibbs states, the von Neumann entropy reduces to the standard thermodynamic entropy $S = (U - F)/T$.']** This definition coincides with the conventional definition of entropy in thermal physics:

$$-\frac{\partial F}{\partial T} = \frac{1}{T}\Big(-\frac{\partial}{\partial \beta}\ln Z_\beta\Big) - \big(-k_{\rm B}\ln Z_\beta\big) = \frac{U-F}{T}. \tag{71}$$

We can derive the same expression using Gibbs states:

$$\begin{aligned}
S =& -k_{\rm B}\mathrm{Tr}\big(\rho\ln\rho\big) = -k_{\rm B}\mathrm{Tr}\big(\rho(-\beta\hat{H} - \ln Z_\beta)\big) \\
=& \frac{\mathrm{Tr}(\rho\hat{H}) - (-k_{\rm B}T\ln Z_\beta)}{T} = \frac{U-F}{T}.
\end{aligned} \tag{72}$$

Using our previous expressions for U and F, we have:

$$\begin{aligned}
U - F &= E_0 + \sum_{\mathbf{k},\mu} E_{\mathbf{k},\mu} n_\mu(\mathbf{k}) - \left(E_0 + \frac{1}{\beta}\sum_{\mathbf{k},\mu}\ln\left(1 - e^{-\beta E_{\mathbf{k},\mu}}\right)\right) \\
&= \sum_{\mathbf{k},\mu} E_{\mathbf{k},\mu} n_\mu(\mathbf{k}) - \frac{1}{\beta}\sum_{\mathbf{k},\mu}\ln\left(1 - e^{-\beta E_{\mathbf{k},\mu}}\right).
\end{aligned} \tag{73}$$

Since $S = k_{\rm B}\beta(U - F)$ (noting $\beta = \frac{1}{k_B T}$ and adjusting units appropriately):

$$S = k_{\rm B}\sum_{\mathbf{k},\mu}\beta E_{\mathbf{k},\mu} n_\mu(\mathbf{k}) - k_{\rm B}\sum_{\mathbf{k},\mu}\ln\left(1 - e^{-\beta E_{\mathbf{k},\mu}}\right). \tag{74}$$

Now, we can rewrite this expression using the properties of the Bose-Einstein distribution function $n_{\mathbf{k},\mu}$. From the definition $n_{\mathbf{k},\mu} = \frac{1}{e^{\beta E_{\mathbf{k},\mu}} - 1}$, we can derive:

$$\beta E_{\mathbf{k},\mu} = \ln\Big(\frac{1 + n_\mu(\mathbf{k})}{n_\mu(\mathbf{k})}\Big), \quad \text{and} \quad \ln\left(1 - e^{-\beta E_{\mathbf{k},\mu}}\right) = -\ln\big(1 + n_\mu(\mathbf{k})\big), \tag{75}$$

Substituting these relations into our entropy expression, we obtain:

$$S = k_{\mathrm{B}} \sum_{\mathbf{k},\mu} \left[(1 + n_\mu(\mathbf{k})) \ln(1 + n_\mu(\mathbf{k})) - n_\mu(\mathbf{k}) \ln n_\mu(\mathbf{k}) \right] \tag{76}$$

This is the standard entropy formula for a system of non-interacting bosons, which emerges naturally from our linear spin wave theory.

E. Specific Heat

The specific heat C is defined as the temperature derivative of the internal energy:

$$C = \frac{\partial U}{\partial T} = \sum_{\mathbf{k},\mu} E_{\mathbf{k},\mu} \frac{\partial n_\mu(\mathbf{k})}{\partial T}, \tag{77}$$

where $n_\mu(\mathbf{k})$ is the Bose-Einstein distribution for magnons μ. The derivative of the distribution with respect to temperature is

$$\frac{\partial n_\mu(\mathbf{k})}{\partial T} = \frac{\partial n_\mu(\mathbf{k})}{\partial \beta} \frac{\partial \beta}{\partial T} = \Big(- \frac{E_{\mathbf{k},\mu} e^{\beta E_{\mathbf{k},\mu}}}{\left(e^{\beta E_{\mathbf{k},\mu}} - 1\right)^2} \Big) \Big(- \frac{\beta}{T} \Big) = \frac{\beta}{T} \frac{E_{\mathbf{k},\mu} e^{\beta E_{\mathbf{k},\mu}}}{\left(e^{\beta E_{\mathbf{k},\mu}} - 1\right)^2}. \tag{78}$$

Therefore, the specific heat is given by:

$$\begin{aligned} C =& \sum_{\mathbf{k},\mu} E_{\mathbf{k},\mu} \frac{\partial n_\mu(\mathbf{k})}{\partial T} = \frac{\beta}{T} \sum_{\mathbf{k},\mu} \frac{E^2_{\mathbf{k},\mu} e^{\beta E_{\mathbf{k},\mu}}}{\left(e^{\beta E_{\mathbf{k},\mu}} - 1\right)^2} = k_{\mathrm{B}} \sum_{\mathbf{k},\mu} \frac{(\beta E_{\mathbf{k},\mu})^2 e^{\beta E_{\mathbf{k},\mu}}}{\left(e^{\beta E_{\mathbf{k},\mu}} - 1\right)^2} = k_{\mathrm{B}} \sum_{\mathbf{k},\mu} \frac{(\beta E_{\mathbf{k},\mu})^2}{\left(e^{\beta E_{\mathbf{k},\mu}/2} - e^{-\beta E_{\mathbf{k},\mu}/2}\right)^2} \\ =& k_{\mathrm{B}} \sum_{\mathbf{k},\mu} \frac{(\beta E_{\mathbf{k},\mu})^2}{4 \sinh^2(\beta E_{\mathbf{k},\mu}/2)} \end{aligned} \tag{79}$$

This expression represents the specific heat contribution from magnon excitations in the system, which dominates the low-temperature thermal properties of magnetic insulators.

F. Number and Spin moment from Correlation Matrix

Finally, we discuss the number expectation and spin moment. Linear spin-wave theory gives the leading correction of the sublattice magnetization. The spin moments are

$$\langle \hat{\mathbf{S}}_\mu \rangle \equiv \frac{1}{L} \sum_i \langle \hat{\mathbf{S}}_{i,\mu} \rangle \tag{80}$$

where the spin moment at site $i = (i, \mu)$ can be obtained from $\langle \hat{\mathbf{S}}_{i,\mu} \rangle = \mathbf{R}_{i,\mu} \langle \widetilde{\mathbf{S}}_{i,\mu} \rangle$ and $\widetilde{\mathbf{S}}_{i,\mu} = (0, 0, \widetilde{S}^0_{i,\mu})$. This is the spin moment in the reference frame. The average sublattice magnetization along the magnetization axis $\widetilde{S}^0_{i,\mu}$ is given

$$\frac{1}{L} \sum_i \langle \widetilde{S}^0_{i,\mu} \rangle = \frac{1}{L} \sum_i \langle S_\mu - \hat{n}_{i,\mu} \rangle = S_\mu - \frac{1}{L} \sum_{\mathbf{k} \in \mathrm{FBZ}} \langle \hat{n}_{\mathbf{k},\mu} \rangle = S_\mu - n_\mu,$$

where i runs over the lattice, μ denotes the sublattice index, and S_μ is the spin moment of the spin. For example, for the spin 1/2 system, $S_\mu = 1/2$.

$$n_\mu \equiv \frac{1}{L} \sum_{\mathbf{k}} \langle \hat{n}_{\mathbf{k},\mu} \rangle = \frac{1}{L} \sum_{\mathbf{k} \in \mathrm{FBZ}} = \langle b^\dagger_{\mathbf{k},\mu} b_{\mathbf{k},\mu} \rangle \tag{81}$$

To proceed, it is useful to introduce the correlation matrix. The correlation matrix is defined as the expectation of outer product of the BdG vector $\Psi_{\mathbf{k}}$.

$$\langle b^\dagger_{\mathbf{k},\mu} b_{\mathbf{k},\mu} \rangle = \left[\langle \Psi_{\mathbf{k}} \Psi^\dagger_{\mathbf{k}} \rangle \right]_{m_s+\mu, m_s+\mu} = \left[\left\langle \begin{pmatrix} \boldsymbol{b}_{\mathbf{k}} \\ \boldsymbol{b}^\dagger_{-\mathbf{k}} \end{pmatrix} \begin{pmatrix} \boldsymbol{b}^\dagger_{\mathbf{k}} & \boldsymbol{b}_{-\mathbf{k}} \end{pmatrix} \right\rangle \right]_{m_s+\mu, m_s+\mu} = \left[\begin{pmatrix} \langle \boldsymbol{b}_{\mathbf{k}} \boldsymbol{b}^\dagger_{\mathbf{k}} \rangle & \langle \boldsymbol{b}_{\mathbf{k}} \boldsymbol{b}_{-\mathbf{k}} \rangle \\ \langle \boldsymbol{b}^\dagger_{-\mathbf{k}} \boldsymbol{b}^\dagger_{\mathbf{k}} \rangle & \langle \boldsymbol{b}^\dagger_{-\mathbf{k}} \boldsymbol{b}_{-\mathbf{k}} \rangle \end{pmatrix} \right]_{m_s+\mu, m_s+\mu} \tag{82}$$

where $\langle \hat{\boldsymbol{b}}^{(\dagger)}_{\pm\mathbf{k}} \hat{\boldsymbol{b}}^{(\dagger)}_{\pm\mathbf{k}} \rangle$ denotes the block matrix whose elements is the expectation for two bosonic operator $(\hat{\boldsymbol{b}}^{(\dagger)}_{\pm\mathbf{k}} \hat{\boldsymbol{b}}^{(\dagger)}_{\pm\mathbf{k}})$. The expectation value is calculated for the state of the interest and our main interests are the ground states and the Gibbs states.

The bosonic operator $(\hat{b}_{\mathbf{k}\sigma}, \hat{b}^{\dagger}_{\mathbf{k}\sigma})$ at $\mathbf{k}$ are related to the eigenmodes $(\hat{\beta}_{\mathbf{k}\sigma}, \hat{\beta}^{\dagger}_{\mathbf{k}\sigma})$ at momentum $\mathbf{k}$ through the para-unitary $\mathsf{T}_{\mathbf{k}}$.

$$\Psi_{\mathbf{k}} = \mathsf{T}_{\mathbf{k}} \widetilde{\Psi}_{\mathbf{k}} : \qquad \begin{pmatrix} \hat{\boldsymbol{b}}_{\mathbf{k}} \\ \hat{\boldsymbol{b}}^{\dagger}_{-\mathbf{k}} \end{pmatrix} = \begin{pmatrix} \mathsf{P}_{\mathbf{k}} & \mathsf{Q}_{-\mathbf{k}} \\ \mathsf{Q}^{*}_{\mathbf{k}} & \mathsf{P}^{*}_{-\mathbf{k}} \end{pmatrix} \begin{pmatrix} \hat{\boldsymbol{\beta}}_{\mathbf{k}} \\ \hat{\boldsymbol{\beta}}^{\dagger}_{-\mathbf{k}} \end{pmatrix}, \tag{83}$$

where $\hat{\boldsymbol{\beta}}^{T}_{\mathbf{k}} = (\hat{\beta}_{\mathbf{k},a}, \hat{\beta}_{\mathbf{k},b}, \cdots)$. Then, the correlation matrix can be obtained using unitary transformation.

$$\big\langle \Psi_{\mathbf{k}} \Psi^{\dagger}_{\mathbf{k}} \big\rangle = \big\langle \mathsf{T}_{\mathbf{k}} \widetilde{\Psi}_{\mathbf{k}} \widetilde{\Psi}^{\dagger}_{\mathbf{k}} \mathsf{T}^{\dagger}_{\mathbf{k}} \big\rangle = \mathsf{T}_{\mathbf{k}} \big\langle \widetilde{\Psi}_{\mathbf{k}} \widetilde{\Psi}^{\dagger}_{\mathbf{k}} \big\rangle \mathsf{T}^{\dagger}_{\mathbf{k}} = \mathsf{T}_{\mathbf{k}} \begin{pmatrix} \langle \hat{\boldsymbol{\beta}}_{\mathbf{k}} \hat{\boldsymbol{\beta}}^{\dagger}_{\mathbf{k}} \rangle & \langle \hat{\boldsymbol{\beta}}_{\mathbf{k}} \hat{\boldsymbol{\beta}}_{-\mathbf{k}} \rangle \\ \langle \hat{\boldsymbol{\beta}}^{\dagger}_{\mathbf{k}} \hat{\boldsymbol{\beta}}^{\dagger}_{-\mathbf{k}} \rangle & \langle \hat{\boldsymbol{\beta}}^{\dagger}_{-\mathbf{k}} \hat{\boldsymbol{\beta}}_{-\mathbf{k}} \rangle \end{pmatrix} \mathsf{T}^{\dagger}_{\mathbf{k}} = \mathsf{T}_{\mathbf{k}} \mathsf{N}_{\mathbf{k}} \mathsf{T}^{\dagger}_{\mathbf{k}}, \tag{84}$$

where $\mathsf{N}_{\mathbf{k}}$ is the $2m_s \times 2m_s$ Block matrix, For the Gibbs states, all elements for the correlation matrices can be obtained from the partition function.

$$\begin{aligned} \big[\langle \hat{\boldsymbol{\beta}}_{\mathbf{k}} \hat{\boldsymbol{\beta}}^{\dagger}_{\mathbf{k}} \rangle\big]_{\mu\nu} &= \delta_{\mu\nu}\big(1 + n_{\mu}(\mathbf{k})\big), \qquad & \big[\langle \hat{\boldsymbol{\beta}}^{\dagger}_{-\mathbf{k}} \hat{\boldsymbol{\beta}}^{\dagger}_{\mathbf{k}} \rangle\big]_{\mu\nu} &= 0, \\ \big[\langle \hat{\boldsymbol{\beta}}^{\dagger}_{-\mathbf{k}} \hat{\boldsymbol{\beta}}_{-\mathbf{k}} \rangle\big]_{\mu\nu} &= \delta_{\mu\nu} n_{\mu}(-\mathbf{k}), & \big[\langle \hat{\boldsymbol{\beta}}_{\mathbf{k}} \hat{\boldsymbol{\beta}}_{-\mathbf{k}} \rangle\big]_{\mu\nu} &= 0. \end{aligned} \tag{85}$$

where $n_{\mu}(\mathbf{k}) = 1/(e^{\beta E_{\mathbf{k}}} - 1)$ is the Bose-Einstein distribution at energy $E_{\mathbf{k}}$.

IV. CORRELATIONS IN LINEAR SPIN WAVE THEORY

In this section, we discuss the physical quantities related to spin-spin correlation functions. Specifically, from real-time (or dynamical) spin-spin correlation function to structure factor and spectral functions, these are quantities of fundamental interest in understanding magnetic excitations. **[REVIEW: B20: Split run-on sentence: 'We cover the real-time correlation, structure factors, and spectral function. These are of fundamental interest for magnetic excitations.'] [REVIEW: C17: Condense intro itemize to brief roadmap; move detailed definitions to respective subsections.]**

- **Real-time correlation function:** The real-time spin-spin correlation function is a 3×3 matrix as a function of momentum and time:

$$\mathcal{C}^{\alpha\beta}(\mathbf{k},t) = \frac{1}{L}\sum_{i,j} e^{-i\mathbf{k}\cdot(\mathbf{r}_i-\mathbf{r}_j)}\langle \hat{S}_i^\alpha(t)\hat{S}_j^\beta(0)\rangle, \tag{86}$$

  where $\alpha,\beta = x,y,z$ or $\pm,0$ and L is the number of lattice sites for spins.

  - Note that the real-time spin-spin correlation function is equivalently defined as

$$\mathcal{C}^{\alpha\beta}(\mathbf{k},t) = \langle \hat{S}^\alpha_{+\mathbf{k}}(t)\hat{S}^\beta_{-\mathbf{k}}(0)\rangle, \tag{87}$$

    by employing the spin Fourier transformation $S^\alpha_{\mathbf{k}} = \frac{1}{\sqrt{N}}\sum_i e^{-i\mathbf{k}\cdot\mathbf{r}_i}S_i^\alpha$.

  - The spin-spin correlation function under complex conjugation is

$$[\mathcal{C}^{\alpha\beta}(\mathbf{k},t)]^* = \langle [S^\alpha_{\mathbf{k}}(t)S^\beta_{-\mathbf{k}}(0)]^\dagger\rangle = \langle (S^\beta_{-\mathbf{k}}(0))^\dagger (S^\alpha_{\mathbf{k}}(t))^\dagger\rangle. \tag{88}$$

    At this point, the spin-spin correlation transforms differently depending on indices α,β:

$$\langle (S^\beta_{-\mathbf{k}}(0))^\dagger (S^\alpha_{\mathbf{k}}(t))^\dagger\rangle = \begin{cases} \langle S^{\bar\beta}_{\mathbf{k}}(0)S^{\bar\alpha}_{-\mathbf{k}}(t)\rangle = \langle S^{\bar\beta}_{\mathbf{k}}(-t)S^{\bar\alpha}_{-\mathbf{k}}(0)\rangle = \mathcal{C}^{\bar\beta\bar\alpha}(-\mathbf{k},-t), & \text{if } \alpha,\beta=\pm,0, \\ \langle S^{\beta}_{\mathbf{k}}(0)S^{\alpha}_{-\mathbf{k}}(t)\rangle = \langle S^{\beta}_{\mathbf{k}}(-t)S^{\alpha}_{-\mathbf{k}}(0)\rangle = \mathcal{C}^{\beta\alpha}(-\mathbf{k},-t), & \text{if } \alpha,\beta=x,y,z. \end{cases} \tag{89}$$

    where in the second equality we use the identity $\langle \hat{A}(0)\hat{B}(t)\rangle = \langle \hat{A}(-t)\hat{B}(0)\rangle$ for thermal states $\rho = e^{-\beta H}/Z$:

$$\langle \hat{A}(0)\hat{B}(t)\rangle = \mathrm{Tr}[\rho\hat{A}\hat{U}_t^\dagger\hat{B}\hat{U}_t] = \mathrm{Tr}[\rho\hat{U}_t\hat{A}\hat{U}_t^\dagger\hat{B}(0)] = \langle \hat{A}(-t)\hat{B}(0)\rangle. \tag{90}$$

- **Static structure factor:** The static structure factor is defined as

$$\mathcal{S}^{\alpha\beta}(\mathbf{k}) \equiv \frac{1}{N}\sum_{i,j} e^{-i\mathbf{k}\cdot(\mathbf{r}_i-\mathbf{r}_j)}\langle \hat{S}_i^\alpha\hat{S}_j^\beta\rangle = \langle \hat{S}^\alpha_{+\mathbf{k}}\hat{S}^\beta_{-\mathbf{k}}\rangle = \mathcal{C}^{\alpha\beta}(\mathbf{k},0). \tag{91}$$

  This equals the equal-time spin-spin correlation function in momentum space, and can also be viewed as its time-average.

- **Dynamical structure factor:** The dynamic structure factor is defined as

$$\mathcal{S}^{\alpha\beta}(\mathbf{k},\omega) \equiv \frac{1}{2\pi}\int_{-\infty}^{\infty} dt\,\mathcal{C}^{\alpha\beta}(\mathbf{k},t)e^{i\omega t}. \tag{92}$$

  This quantity is directly related to the scattering cross-section in neutron scattering experiments.

- **Real space correlation function in the reference frame:** The spin-spin correlation in the local reference frame is:

$$\langle \widetilde{S}_i^\alpha\widetilde{S}_j^\beta\rangle, \tag{93}$$

  where $\widetilde{S}_i^\alpha$ represents the spin component in the local frame.

- **Spectral function:** The spectral function is defined as:

$$\mathcal{A}^{\alpha\beta}(\mathbf{k},\omega) = -\frac{1}{\pi}\mathrm{Im}[G_{\mathrm{R}}^{\alpha\beta}(\mathbf{k},\omega)], \tag{94}$$

where $G_{\mathrm{R}}^{\alpha\beta}(\mathbf{k},\omega)$ is the retarded Green's function:

$$G_{\mathrm{R}}^{\alpha\beta}(\mathbf{k},\omega) \equiv \frac{1}{i}\int_{-\infty}^{\infty} dt\, e^{i\omega t}\theta(t)\big\langle[S_{\mathbf{k}}^{\alpha}(t), S_{-\mathbf{k}}^{\beta}(0)]\big\rangle. \tag{95}$$

The spectral function provides information about the density of states of the magnetic excitations.

A. Real-Time (Dynamical) Spin-Spin Correlation Function

In this section, we derive the spin-spin correlation function in linear spin-wave theory (LSWT). The real-time spin-spin correlation function is a 3×3 matrix as a function of momentum and time:

$$\mathcal{C}^{\alpha\beta}(\mathbf{k},t) = \frac{1}{L}\sum_{i,j} e^{-i\mathbf{k}\cdot(\mathbf{r}_i-\mathbf{r}_j)}\langle \hat{S}_i^{\alpha}(t)\hat{S}_j^{\beta}(0)\rangle, \tag{96}$$

where $\alpha,\beta = x,y,z$ or $\pm,0$ and L is the number of lattice sites for spins. To proceed, we express the α-component of the spin operator as

$$\hat{S}_i^{\alpha} = \hat{\mathbf{e}}^{\alpha}\cdot\left(\frac{\widetilde{S}_i^{+}\hat{\mathbf{e}}_i^{-}+\widetilde{S}_i^{-}\hat{\mathbf{e}}_i^{+}}{\sqrt{2}} + \widetilde{S}_i^{0}\hat{\mathbf{e}}_i^{0}\right) = \frac{1}{\sqrt{2}}\left(\bar{u}_i^{\alpha}\widetilde{S}_i^{+} + u_i^{\alpha}\widetilde{S}_i^{-}\right) + v_i^{\alpha}\widetilde{S}_i^{0}, \tag{97}$$

where $u_i^{\alpha} = \hat{\mathbf{e}}^{\alpha}\cdot\hat{\mathbf{e}}_i^{+}$, $\bar{u}_i^{\alpha} = (u_i^{\alpha})^*$, and $v_i^{\alpha} = \hat{\mathbf{e}}^{\alpha}\cdot\hat{\mathbf{e}}_i^{0}$ are components of the spin basis vectors. When the index α are $\pm,0$, then the $(u_i^{\alpha},\bar{u}_i^{\alpha},v_i^{\alpha})$ can be obtained from $\mathbf{Q}$:

$$[\mathbf{Q}]_{\alpha\beta} = \hat{\mathbf{e}}^{\alpha}\cdot\hat{\mathbf{e}}_i^{\beta} = \hat{\mathbf{e}}^{\alpha}\cdot\mathbf{R}_i\hat{\mathbf{e}}^{\beta}: \qquad \mathbf{Q} = \begin{pmatrix} u_i^{-} & \bar{u}_i^{-} & v_i^{-} \\ u_i^{+} & \bar{u}_i^{+} & v_i^{+} \\ u_i^{0} & \bar{u}_i^{0} & v_i^{0}\end{pmatrix} = \begin{pmatrix}(\hat{\mathbf{e}}^{-})^{\mathrm{T}} \\ (\hat{\mathbf{e}}^{+})^{\mathrm{T}} \\ (\hat{\mathbf{e}}^{0})^{\mathrm{T}}\end{pmatrix}\mathbf{R}_i\begin{pmatrix}\hat{\mathbf{e}}^{+} & \hat{\mathbf{e}}^{-} & \hat{\mathbf{e}}^{0}\end{pmatrix} = \mathbf{C}^{\dagger}\mathbf{R}_i\mathbf{C}, \tag{98}$$

If the indices α are x,y,z, $\mathbf{U}_i = \mathbf{R}_i\mathbf{C}$. Substituting into the spin-spin correlation function, we obtain

$$\begin{aligned}\langle S_i^{\alpha}(t)S_j^{\beta}(0)\rangle = &\frac{1}{2}\left\langle\left(\bar{u}_i^{\alpha}\widetilde{S}_i^{+}(t)+u_i^{\alpha}\widetilde{S}_i^{-}(t)\right)\left(\bar{u}_j^{\beta}\widetilde{S}_j^{+}(0)+u_j^{\beta}\widetilde{S}_j^{-}(0)\right)\right\rangle + v_i^{\alpha}v_j^{\beta}\langle\widetilde{S}_i^{0}(t)\widetilde{S}_j^{0}(0)\rangle \\ &+\frac{1}{\sqrt{2}}\left[v_i^{\alpha}\left\langle\widetilde{S}_i^{0}(t)\left(\bar{u}_j^{\beta}\widetilde{S}_j^{+}(0)+u_j^{\beta}\widetilde{S}_j^{-}(0)\right)\right\rangle + v_j^{\beta}\left\langle\left(\bar{u}_i^{\alpha}\widetilde{S}_i^{+}(t)+u_i^{\alpha}\widetilde{S}_i^{-}(t)\right)\widetilde{S}_j^{0}(0)\right\rangle\right].\end{aligned} \tag{99}$$

In LSWT, which is a quadratic bosonic system, odd-order bosonic correlation functions vanish, so the terms in the second line are zero. Thus,

$$\langle S_i^{\alpha}(t)S_j^{\beta}(0)\rangle = \frac{1}{2}\left\langle\left(\bar{u}_i^{\alpha}\widetilde{S}_i^{+}(t)+u_i^{\alpha}\widetilde{S}_i^{-}(t)\right)\left(\bar{u}_j^{\beta}\widetilde{S}_j^{+}(0)+u_j^{\beta}\widetilde{S}_j^{-}(0)\right)\right\rangle + v_i^{\alpha}v_j^{\beta}\langle\widetilde{S}_i^{0}(t)\widetilde{S}_j^{0}(0)\rangle \tag{100}$$

$$= \frac{1}{2}\sum_{m,n}\left[\begin{pmatrix}\bar{u}_i^{\alpha} & 0 \\ 0 & u_i^{\alpha}\end{pmatrix}\begin{pmatrix}\langle\widetilde{S}_i^{+}(t)\widetilde{S}_j^{-}(0)\rangle & \langle\widetilde{S}_i^{+}(t)\widetilde{S}_j^{+}(0)\rangle \\ \langle\widetilde{S}_i^{-}(t)\widetilde{S}_j^{-}(0)\rangle & \langle\widetilde{S}_i^{-}(t)\widetilde{S}_j^{+}(0)\rangle\end{pmatrix}\begin{pmatrix}u_j^{\beta} & 0 \\ 0 & \bar{u}_j^{\beta}\end{pmatrix}\right]_{m,n} + v_i^{\alpha}v_j^{\beta}\langle\widetilde{S}_i^{0}(t)\widetilde{S}_j^{0}(0)\rangle. \tag{101}$$

Now, we define two matrix quantities: U_i^{α}, which maps the spin operators from the reference frame to the laboratory frame, and the correlation matrix between spin ladder operators (or transversal fluctuations).

$$\mathsf{U}_i^{\alpha} \equiv \begin{pmatrix}\bar{u}_i^{\alpha} & 0 \\ 0 & u_i^{\alpha}\end{pmatrix}, \qquad \text{and} \qquad \big\langle\widetilde{\mathbf{S}}_i^{\pm}(t)\widetilde{\mathbf{S}}_j^{\mp}(0)\big\rangle = \begin{pmatrix}\langle\widetilde{S}_i^{+}(t)\widetilde{S}_j^{-}(0)\rangle & \langle\widetilde{S}_i^{+}(t)\widetilde{S}_j^{+}(0)\rangle \\ \langle\widetilde{S}_i^{-}(t)\widetilde{S}_j^{-}(0)\rangle & \langle\widetilde{S}_i^{-}(t)\widetilde{S}_j^{+}(0)\rangle\end{pmatrix} \tag{102}$$

The Holstein-Primakoff (HP) operators are

$$\widetilde{S}_i^+ = \sqrt{2S_i}\left(1-\frac{\hat{n}_i}{2S_i}\right)^{1/2}\hat{b}_i, \qquad \widetilde{S}_i^- = \sqrt{2S_i}\hat{b}_i^\dagger\left(1-\frac{\hat{n}_i}{2S_i}\right)^{1/2}, \qquad \widetilde{S}_i^0 = S_i - \hat{n}_i, \tag{103}$$

with $\hat{n}_i = \hat{b}_i^\dagger \hat{b}_i$ being the magnon number operator. We approximate $\widetilde{S}_i^+ \approx \sqrt{2S_i}\hat{b}_i$ and $\widetilde{S}_i^- \approx \sqrt{2S_i}\hat{b}_i^\dagger$, yielding

$$\langle S_i^\alpha(t) S_j^\beta(0)\rangle = \sqrt{S_i S_j}\sum_{m,n}\left[\mathsf{U}_i^\alpha \left\langle \Psi_i(t)\Psi_j^\dagger(0)\right\rangle \mathsf{U}_j^\beta\right]_{m,n} + v_i^\alpha v_j^\beta \left(S_i S_j - S_i\langle \hat{n}_j(0)\rangle - S_j\langle \hat{n}_i(t)\rangle\right), \tag{104}$$

where $\left\langle \Psi_i(t)\Psi_j^\dagger(0)\right\rangle$ is the matrix form of the two-point correlation function, defined as

$$\left\langle \Psi_i(t)\Psi_j^\dagger(0)\right\rangle \equiv \begin{pmatrix} \langle \hat{b}_i(t)\hat{b}_j^\dagger(0)\rangle & \langle \hat{b}_i(t)\hat{b}_j(0)\rangle \\ \langle \hat{b}_i^\dagger(t)\hat{b}_j^\dagger(0)\rangle & \langle \hat{b}_i^\dagger(t)\hat{b}_j(0)\rangle \end{pmatrix} = \frac{1}{\sqrt{S_i S_j}}\left\langle \widetilde{\mathbf{S}}_i^\pm(t)\widetilde{\mathbf{S}}_j^\mp(0)\right\rangle \tag{105}$$

The expectation value of the number operator satisfies $\langle \hat{n}_i(t)\rangle = \langle \hat{n}_i(0)\rangle$ because, for a time-independent Hamiltonian $\hat{H}$,

$$\langle \hat{A}(t)\rangle = \begin{cases} \langle \psi_n | e^{i\hat{H}t}\hat{A}e^{-i\hat{H}t}|\psi_n\rangle = \langle \hat{A}\rangle, & \text{if } \hat{H}|\psi_n\rangle = E_n|\psi_n\rangle, \\ \mathrm{Tr}[\rho \hat{U}^\dagger(t)\hat{A}\hat{U}(t)] = \mathrm{Tr}[\rho\hat{A}], & \text{if } \rho = e^{-\beta\hat{H}}/Z, \end{cases} \tag{106}$$

where $\hat{U}(t) = e^{-i\hat{H}t}$ is the time-evolution operator and $\rho = e^{-\beta\hat{H}}/Z$ is the thermal density matrix. In the Gibbs state, this follows from the commutation of ρ with $\hat{U}(t)$ and the cyclic property of the trace.

Sublattice Spin-Spin Correlation Function

Given L spins (or magnetic ions), there could be m_s magnetic sublattices and N unit cells with $L = m_s N$. In this case, we introduce a composite index notation: $I = (i,\mu)$ and $J = (j,\nu)$, where $\mu,\nu = a,b,\dots$ denote the magnetic sublattices and i,j represent the unit cell indices. The position in the lattice is given by $\mathbf{R}_I = \mathbf{r}_i + \boldsymbol{\delta}_\mu$, where $\mathbf{r}_i$ is the position of the i-th unit cell and $\boldsymbol{\delta}_\mu$ is the position of the μ-th sublattice within the unit cell.

Explicitly, for a spin at site $I = (i,\mu)$, its position vector is

$$\mathbf{R}_I = \mathbf{R}_{i\mu} = \mathbf{r}_i + \boldsymbol{\delta}_\mu. \tag{107}$$

To compute the real-time spin-spin correlation function in momentum space using Linear Spin-Wave Theory (LSWT), we express it in terms of magnetic sublattices:

$$\mathcal{C}^{\alpha\beta}(\mathbf{k},t) = \frac{1}{Nm_s}\sum_{i\mu,j\nu} e^{-i\mathbf{k}\cdot[(\mathbf{r}_i+\boldsymbol{\delta}_\mu)-(\mathbf{r}_j+\boldsymbol{\delta}_\nu)]}\langle \hat{S}_{i\mu}^\alpha(t)\hat{S}_{j\nu}^\beta(0)\rangle \tag{108}$$

$$= \frac{1}{m_s}\sum_{\mu,\nu} e^{-i\mathbf{k}\cdot(\boldsymbol{\delta}_\mu-\boldsymbol{\delta}_\nu)}\left[\frac{1}{N}\sum_{i,j} e^{-i\mathbf{k}\cdot(\mathbf{r}_i-\mathbf{r}_j)}\langle \hat{S}_{i\mu}^\alpha(t)\hat{S}_{j\nu}^\beta(0)\rangle\right] \tag{109}$$

This allows us to express the total correlation function as an average over sublattice correlation functions:

$$\mathcal{C}^{\alpha\beta}(\mathbf{k},t) = \frac{1}{m_s}\sum_{\mu,\nu=a,b,\dots}\mathcal{C}_{\mu\nu}^{\alpha\beta}(\mathbf{k},t), \tag{110}$$

where m_s is the number of magnetic sublattices in the unit cell, and μ,ν denote sublattice indices.

The key insight of this section is to demonstrate that the sublattice real-time spin-spin correlation function $\mathcal{C}_{\mu\nu}^{\alpha\beta}(\mathbf{k},t)$ can be obtained from a partial sum of the $2m_s \times 2m_s$ matrix $\mathsf{C}^{\alpha\beta}(\mathbf{k},t)$:

$$\begin{aligned}\mathcal{C}_{\mu\nu}^{\alpha\beta}(\mathbf{k},t) &= \sum_{(m,n)_{\mu,\nu}}\left[\mathsf{C}^{\alpha\beta}(\mathbf{k},t)\right]_{m,n} \\ &= \left[\mathsf{C}^{\alpha\beta}(\mathbf{k},t)\right]_{\mu,\nu} + \left[\mathsf{C}^{\alpha\beta}(\mathbf{k},t)\right]_{m_s+\mu,\nu} + \left[\mathsf{C}^{\alpha\beta}(\mathbf{k},t)\right]_{\mu,m_s+\nu} + \left[\mathsf{C}^{\alpha\beta}(\mathbf{k},t)\right]_{m_s+\mu,m_s+\nu},\end{aligned} \tag{111}$$

where the index set $(m,n)_{\mu,\nu} = \{(\mu,\nu),(m_s+\mu,\nu),(\mu,m_s+\nu),(m_s+\mu,m_s+\nu)\}$ represents all possible combinations of indices in the extended Nambu space. The matrix $\mathsf{C}^{\alpha\beta}$ is defined as

$$\boxed{\mathsf{C}^{\alpha\beta}(\mathbf{k},t) = \mathsf{R}^{\alpha}_{\mathbf{k}}\mathsf{T}_{\mathbf{k}}\mathsf{N}_{\mathbf{k}}(t)\mathsf{T}^{\dagger}_{\mathbf{k}}[\mathsf{R}^{\beta}_{\mathbf{k}}]^{\dagger},} \tag{112}$$

where $\mathsf{T}_{\mathbf{k}}$ is a para-unitary matrix that diagonalizes the Hamiltonian matrix, satisfying $\mathsf{T}^{\dagger}_{\mathbf{k}}\mathsf{H}_{\mathbf{k}}\mathsf{T}_{\mathbf{k}} = \mathsf{E}_{\mathbf{k}}$, and $\mathsf{N}_{\mathbf{k}}(t)$ is the time-dependent correlation matrix in the magnon basis.

The matrix $\mathsf{R}^{\alpha}_{\mathbf{k}}$ is a $2m_s \times 2m_s$ block-diagonal matrix that encodes both the spin components and sublattice structure. It is defined as $\mathsf{R}^{\alpha}_{\mathbf{k}} = \mathsf{U}^{\alpha}\mathsf{S}_{\mathbf{k}}$, which can be expressed explicitly as:

$$\mathsf{R}^{\alpha}_{\mathbf{k}} = \mathsf{U}^{\alpha}\mathsf{S}_{\mathbf{k}} = \begin{pmatrix} \bar{\mathsf{u}}^{\alpha}\mathsf{s}_{\mathbf{k}} & \mathbf{0} \\ \mathbf{0} & \mathsf{u}^{\alpha}\mathsf{s}_{\mathbf{k}} \end{pmatrix}, \tag{113}$$

where the diagonal matrices s and u^{α} are defined as:

$$\begin{cases} \mathsf{s} = \mathrm{diag}\big[e^{-i\mathbf{k}\cdot\boldsymbol{\delta}_a}\sqrt{S_a}, e^{-i\mathbf{k}\cdot\boldsymbol{\delta}_b}\sqrt{S_b}, \dots, e^{-i\mathbf{k}\cdot\boldsymbol{\delta}_\mu}\sqrt{S_\mu}, \dots\big], \\ \mathsf{u}^{\alpha} = \mathrm{diag}\big[u^{\alpha}_a, u^{\alpha}_b, \dots, u^{\alpha}_\mu, \dots\big], \end{cases} \tag{114}$$

where u^{α} contains the α-component of the spin basis vector $\hat{\mathbf{e}}^{-}$ for each sublattice, i.e., $\mathsf{u}^{\alpha}_{\mu} = [\hat{\mathbf{e}}^{+}_{\mu}]^{\alpha}$, and $\bar{\mathsf{u}}^{\alpha} = [\mathsf{u}^{\alpha}]^{*} = [\mathsf{u}^{\alpha}]^{\dagger}$. The matrix s encodes both the phase factors due to sublattice positions and the square root of spin magnitudes S_μ.

The matrix $\mathsf{N}_{\mathbf{k}}(t)$ encodes the time evolution of magnon correlations and is given by:

$$\mathsf{N}_{\mathbf{k}}(t) = \begin{pmatrix} \langle\hat{\boldsymbol{\beta}}_{\mathbf{k}}(t)\hat{\boldsymbol{\beta}}^{\dagger}_{\mathbf{k}}(0)\rangle & \langle\hat{\boldsymbol{\beta}}_{\mathbf{k}}(t)\hat{\boldsymbol{\beta}}_{-\mathbf{k}}(0)\rangle \\ \langle\hat{\boldsymbol{\beta}}^{\dagger}_{-\mathbf{k}}(t)\hat{\boldsymbol{\beta}}^{\dagger}_{\mathbf{k}}(0)\rangle & \langle\hat{\boldsymbol{\beta}}^{\dagger}_{-\mathbf{k}}(t)\hat{\boldsymbol{\beta}}_{-\mathbf{k}}(0)\rangle \end{pmatrix} = \begin{pmatrix} (\mathsf{I}+\mathsf{n}_{\mathbf{k}})e^{-it\mathsf{E}_{\mathbf{k}}} & \mathbf{0} \\ \mathbf{0} & \mathsf{n}_{-\mathbf{k}}e^{it\mathsf{E}_{-\mathbf{k}}} \end{pmatrix}, \tag{115}$$

where I is the $m_s \times m_s$ identity matrix, $\mathsf{n}_{\mathbf{k}}$ and $\mathsf{E}_{\mathbf{k}}$ are diagonal matrices containing the Bose-Einstein distribution $[\mathsf{n}_{\mathbf{k}}]_{\mu\nu} = \delta_{\mu\nu}/[e^{\beta E_\mu(\mathbf{k})} - 1]$ and the magnon energies $[\mathsf{E}_{\mathbf{k}}]_{\mu\nu} = \delta_{\mu\nu}E_\mu(\mathbf{k})$, respectively. The time dependence arises from the energy eigenstates: $e^{i\hat{H}t}\hat{\beta}_{\mathbf{k},\mu}e^{-i\hat{H}t} = e^{-iE_{\mathbf{k},\mu}t}\hat{\beta}_{\mathbf{k},\mu}$.

To derive this, we assume a spin system with magnetic sublattice structures within the unit cell. The sublattice correlation function is expressed as

$$\mathcal{C}^{\alpha\beta}_{\mu\nu}(\mathbf{k},t) = \frac{1}{N}\sum_{i,j} e^{-i\mathbf{k}\cdot[(\mathbf{r}_i+\delta_\mu)-(\mathbf{r}_j+\delta_\nu)]}\langle S^{\alpha}_{i\mu}(t)S^{\beta}_{j\nu}(0)\rangle \tag{116}$$

$$= e^{-i\mathbf{k}\cdot(\delta_\mu-\delta_\nu)}\frac{1}{N}\sum_{i,j} e^{-i\mathbf{k}\cdot(\mathbf{r}_i-\mathbf{r}_j)}\langle S^{\alpha}_{i\mu}(t)S^{\beta}_{j\nu}(0)\rangle \tag{117}$$

$$= e^{-i\mathbf{k}\cdot(\delta_\mu-\delta_\nu)}\frac{1}{N} \tag{118}$$

$$\times\sum_{i,j} e^{-i\mathbf{k}\cdot(\mathbf{r}_i-\mathbf{r}_j)}\left[\sqrt{S_\mu S_\nu}\sum_{m,n}\Big[\mathsf{U}^{\alpha}_{\mu}\langle\Psi_{i\mu}(t)\Psi^{\dagger}_{j\nu}(0)\rangle\mathsf{U}^{\beta}_{\nu}\Big]_{m,n} + v^{\alpha}_{\mu}v^{\beta}_{\nu}\Big(S_\nu S_\mu - S_\nu\langle\hat{n}_{i\mu}\rangle - S_\mu\langle\hat{n}_{j\nu}\rangle\Big)\right],$$

where the spin operators are transformed into bosonic operators via the Holstein-Primakoff transformation, and the correlation is computed using Fourier transforms. The first term, involving the bosonic correlation matrix, yields

$$\frac{1}{N}\sum_{i,j} e^{-i\mathbf{k}\cdot(\mathbf{r}_i-\mathbf{r}_j)}\big\langle\Psi_i(t)\Psi^{\dagger}_j(0)\big\rangle = \big\langle\Psi_{\mathbf{k}}(t)\Psi^{\dagger}_{\mathbf{k}}(0)\big\rangle, \tag{119a}$$

$$\frac{1}{N}\sum_{\mathbf{k}} e^{+i\mathbf{k}\cdot(\mathbf{r}_i-\mathbf{r}_j)}\big\langle\Psi_{\mathbf{k}}(t)\Psi^{\dagger}_{\mathbf{k}}(0)\big\rangle = \big\langle\Psi_i(t)\Psi^{\dagger}_j(0)\big\rangle. \tag{119b}$$

Similarly, the spin-spin correlation function in real space is obtained as

$$\big\langle\widetilde{\mathbf{S}}^{\pm}_i(t)\widetilde{\mathbf{S}}^{\mp}_j(0)\big\rangle = \frac{1}{N}\sum_{\mathbf{k},\mu,\nu} e^{+i\mathbf{k}\cdot(\mathbf{r}_i-\mathbf{r}_j)}\sqrt{S_\mu S_\nu}e^{i\mathbf{k}\cdot(\delta_\mu-\delta_\nu)}\big\langle\Psi_{\mathbf{k},\mu}(t)\Psi^{\dagger}_{\mathbf{k},\nu}(0)\big\rangle = \frac{1}{N}\sum_{\mathbf{k}} e^{+i\mathbf{k}\cdot(\mathbf{r}_i-\mathbf{r}_j)}\bar{\mathsf{S}}_{\mathbf{k}}\mathsf{T}_{\mathbf{k}}\mathsf{N}_{\mathbf{k}}(t)\mathsf{T}^{\dagger}_{\mathbf{k}}\mathsf{S}_{\mathbf{k}}, \tag{120}$$

derived by applying the Fourier transform to the bosonic operators and computing the correlation matrix elements:

$$\frac{1}{N}\sum_{i,j} e^{-i\mathbf{k}\cdot(\mathbf{r}_i-\mathbf{r}_j)}\big\langle\Psi_{i\mu}(t)\Psi^{\dagger}_{j\nu}(0)\big\rangle = \begin{pmatrix} \big\langle\hat{b}_{+\mathbf{k},\mu}(t)\hat{b}^{\dagger}_{+\mathbf{k},\nu}(0)\big\rangle & \big\langle\hat{b}_{+\mathbf{k},\mu}(t)\hat{b}_{-\mathbf{k},\nu}(0)\big\rangle \\ \big\langle\hat{b}^{\dagger}_{-\mathbf{k},\mu}(t)\hat{b}^{\dagger}_{+\mathbf{k},\nu}(0)\big\rangle & \big\langle\hat{b}_{-\mathbf{k},\mu}(t)\hat{b}^{\dagger}_{-\mathbf{k},\nu}(0)\big\rangle \end{pmatrix} = \mathsf{T}_{\mathbf{k}}\mathsf{N}_{\mathbf{k}}(t)\mathsf{T}^{\dagger}_{\mathbf{k}}, \tag{121}$$

where $\mathsf{T}_{\mathbf{k}}$ and $\mathsf{N}_{\mathbf{k}}(t)$ are defined above. The second term, representing the density operator in momentum space, is

$$\frac{1}{N}\sum_{i,j} e^{-i\mathbf{k}\cdot(\mathbf{r}_i-\mathbf{r}_j)}\langle \hat{n}_{j\nu}\rangle = \Big(\sum_{\kappa\in\mathbf{G}}\delta(\mathbf{k}-\kappa)\Big)\langle \hat{n}_{\mathbf{k},\mu}\rangle, [REVIEW: A13: On discrete lattice,$$

$\delta(k-\kappa)$ should be Kronecker delta $\delta_{k,\kappa}$, not Dirac delta.] (122)

where $\hat{n}_{\mathbf{k},\mu}$, the density fluctuation operator at momentum $\mathbf{k}$, is

$$\hat{n}_{\mathbf{k}} = \frac{1}{N}\sum_i e^{i\mathbf{k}\cdot\mathbf{r}_i}\hat{a}_i^\dagger\hat{a}_i = \frac{1}{N}\sum_{\mathbf{q}}\hat{a}^\dagger_{\mathbf{q}+\mathbf{k}}\hat{a}_{\mathbf{q}},$$

which vanishes for a free system unless $\mathbf{k}=\kappa\in\mathbf{G}$ due to momentum conservation.

The correlation function is expressed in matrix form as

$$\mathcal{C}^{\alpha\beta}_{\mu\nu}(\mathbf{k},t) = e^{-i\mathbf{k}\cdot(\delta_\mu-\delta_\nu)}\left[\sqrt{S_\mu S_\nu}\sum_{m,n}\Big[\mathsf{U}^\alpha_\mu\mathsf{T}_{\mathbf{k}}\mathsf{N}_{\mathbf{k}}(t)\mathsf{T}^\dagger_{\mathbf{k}}\mathsf{U}^\beta_\nu\Big]_{m,n} + v^\alpha_\mu v^\beta_\nu\Big(\sum_{\kappa\in\mathbf{G}}\delta(\mathbf{k}-\kappa)\Big)\Big(S_\nu\langle\hat{n}_{\mathbf{0},\mu}\rangle + S_\mu\langle\hat{n}_{\mathbf{0},\nu}\rangle\Big)\right], \quad (123)$$

where the first term describes time-dependent magnon scattering, and the second term accounts for the static ordered moment reduction due to magnon population. The matrix form is

$$\mathsf{C}^{\alpha\beta}(\mathbf{k},t) = \mathsf{U}^\alpha\mathsf{S}_{\mathbf{k}}\mathsf{T}_{\mathbf{k}}\mathsf{N}_{\mathbf{k}}(t)\mathsf{T}^\dagger_{\mathbf{k}}\mathsf{S}^\dagger_{\mathbf{k}}\mathsf{U}^\beta, \quad (124)$$

with elements computed as

$$\mathsf{C}^{\alpha\beta}(\mathbf{k},t) = \begin{pmatrix}\bar{\mathsf{u}}^\alpha\mathsf{s}_{\mathbf{k}} & \mathbf{0}\\ \mathbf{0} & \mathsf{u}^\alpha\mathsf{s}_{\mathbf{k}}\end{pmatrix}\begin{pmatrix}\langle\mathbf{b}_{\mathbf{k}\mu}(t)\mathbf{b}^\dagger_{\mathbf{k}\nu}(0)\rangle & \langle\mathbf{b}_{\mathbf{k}\mu}(t)\mathbf{b}_{-\mathbf{k}\nu}(0)\rangle\\ \langle\mathbf{b}^\dagger_{-\mathbf{k}\mu}(t)\mathbf{b}^\dagger_{\mathbf{k}\nu}(0)\rangle & \langle\mathbf{b}^\dagger_{-\mathbf{k}\mu}(t)\mathbf{b}_{-\mathbf{k}\nu}(0)\rangle\end{pmatrix}\begin{pmatrix}\bar{\mathsf{s}}_{\mathbf{k}}\mathsf{u}^\beta & \mathbf{0}\\ \mathbf{0} & \bar{\mathsf{s}}_{\mathbf{k}}\bar{\mathsf{u}}^\beta\end{pmatrix}.$$

The elements of $\mathsf{C}^{\alpha\beta}(\mathbf{k},t)$ for indices $(m,n)_{\mu,\nu}$ are

$$\begin{aligned}
[\mathsf{C}^{\alpha\beta}(\mathbf{k},t)]_{(1,1)_{\mu\nu}}:&\quad \Big[\mathsf{s}_{\mathbf{k}}\bar{\mathsf{u}}^\alpha\langle\mathbf{b}_{+\mathbf{k}\mu}(t)\mathbf{b}^\dagger_{+\mathbf{k}\nu}(0)\rangle\mathsf{u}^\beta\bar{\mathsf{s}}_{\mathbf{k}}\Big]_{\mu\nu} = \sqrt{S_\mu S_\nu}e^{-i\mathbf{k}\cdot(\boldsymbol{\delta}_\mu-\boldsymbol{\delta}_\nu)}\big\langle b_{+\mathbf{k}\mu}(t)b^\dagger_{+\mathbf{k}\nu}(0)\big\rangle,\\
[\mathsf{C}^{\alpha\beta}(\mathbf{k},t)]_{(1,2)_{\mu\nu}}:&\quad \Big[\mathsf{s}_{\mathbf{k}}\bar{\mathsf{u}}^\alpha\langle\mathbf{b}_{+\mathbf{k}\mu}(t)\mathbf{b}_{-\mathbf{k}\nu}(0)\rangle\bar{\mathsf{u}}^\beta\bar{\mathsf{s}}_{\mathbf{k}}\Big]_{\mu\nu} = \sqrt{S_\mu S_\nu}e^{-i\mathbf{k}\cdot(\boldsymbol{\delta}_\mu-\boldsymbol{\delta}_\nu)}\big\langle b_{+\mathbf{k}\mu}(t)b_{-\mathbf{k}\nu}(0)\big\rangle,\\
[\mathsf{C}^{\alpha\beta}(\mathbf{k},t)]_{(2,1)_{\mu\nu}}:&\quad \Big[\mathsf{s}_{\mathbf{k}}\mathsf{u}^\alpha\langle\mathbf{b}^\dagger_{-\mathbf{k}\mu}(t)\mathbf{b}^\dagger_{+\mathbf{k}\nu}(0)\rangle\mathsf{u}^\beta\bar{\mathsf{s}}_{\mathbf{k}}\Big]_{\mu\nu} = \sqrt{S_\mu S_\nu}e^{-i\mathbf{k}\cdot(\boldsymbol{\delta}_\mu-\boldsymbol{\delta}_\nu)}\big\langle b^\dagger_{-\mathbf{k}\mu}(t)b^\dagger_{+\mathbf{k}\nu}(0)\big\rangle,\\
[\mathsf{C}^{\alpha\beta}(\mathbf{k},t)]_{(2,2)_{\mu\nu}}:&\quad \Big[\mathsf{s}_{\mathbf{k}}\mathsf{u}^\alpha\langle\mathbf{b}^\dagger_{-\mathbf{k}\mu}(t)\mathbf{b}_{-\mathbf{k}\nu}(0)\rangle\bar{\mathsf{u}}^\beta\bar{\mathsf{s}}_{\mathbf{k}}\Big]_{\mu\nu} = \sqrt{S_\mu S_\nu}e^{-i\mathbf{k}\cdot(\boldsymbol{\delta}_\mu-\boldsymbol{\delta}_\nu)}\big\langle b^\dagger_{-\mathbf{k}\mu}(t)b_{-\mathbf{k}\nu}(0)\big\rangle.
\end{aligned}$$

Summing $\mathsf{C}^{\alpha\beta}(\mathbf{k},t)$ over $(m,n)_{\mu,\nu}$ yields the sublattice spin-spin correlation function

$$\mathcal{C}^{\alpha\beta}_{\mu\nu}(\mathbf{k},t) = \sum_{(m,n)_{\mu,\nu}}[\mathsf{C}^{\alpha\beta}(\mathbf{k},t)]_{m,n}. \quad (125)$$

The spin structure factor is obtained by summing over all sublattices: $\mathcal{S}^{\alpha\beta}(\mathbf{k},t) = \mathcal{C}^{\alpha\beta}(\mathbf{k},t) = \frac{1}{m_s}\sum_{\mu,\nu}\mathcal{C}^{\alpha\beta}_{\mu\nu}(\mathbf{k},t)$.

B. Structure Factor

In this section, we derive the formulas for the static and dynamic structure factors in Linear Spin-Wave Theory.

Static Structure Factor

The static structure factor in momentum space is given by

$$\mathcal{S}^{\alpha\beta}(\mathbf{k}) \equiv \frac{1}{L}\sum_{I,J} e^{-i\mathbf{k}\cdot(\mathbf{r}_I-\mathbf{r}_J)}\langle S^\alpha_I S^\beta_J\rangle = \big\langle S^\alpha_{\mathbf{k}} S^\beta_{-\mathbf{k}}\big\rangle. \quad (126)$$

This quantity $\mathcal{S}^{\alpha\beta}(\mathbf{k})$ is equal to $\mathcal{C}^{\alpha\beta}(\mathbf{k},t)$, which we've found the matrix form in a previous equation:

$$\mathcal{S}^{\alpha\beta}(\mathbf{k}) = \sum_{m,n} \left[\mathsf{U}^{\alpha}\mathsf{S}_{\mathbf{k}}\mathsf{T}_{\mathbf{k}}\mathsf{N}_{\mathbf{k}}\mathsf{T}_{\mathbf{k}}^{\dagger}\mathsf{S}_{\mathbf{k}}^{\dagger}\mathsf{U}^{\beta}\right]_{mn}, \tag{127}$$

where U^{α} and U^{β} are spin projection operators, $\mathsf{S}_{\mathbf{k}}$ and $\mathsf{T}_{\mathbf{k}}$ are transformation matrices, and $\mathsf{N}_{\mathbf{k}}$ accounts for the diagonalized magnon correlation functions. In the reference frame, the structure factor is expressed as

$$\begin{aligned}\widetilde{\mathcal{S}}^{\alpha\beta}(\mathbf{k}) \equiv& \frac{1}{L}\sum_{I,J} e^{-i\mathbf{k}\cdot(\mathbf{r}_I-\mathbf{r}_J)}\langle \widetilde{S}_I^{\alpha}\widetilde{S}_J^{\beta}\rangle = \sum_{\mu,\nu}\left[\begin{pmatrix}\langle \widetilde{S}_{\mathbf{k}}^{+}\widetilde{S}_{-\mathbf{k}}^{-}\rangle & \langle \widetilde{S}_{\mathbf{k}}^{+}\widetilde{S}_{-\mathbf{k}}^{+}\rangle \\ \langle \widetilde{S}_{\mathbf{k}}^{-}\widetilde{S}_{-\mathbf{k}}^{-}\rangle & \langle \widetilde{S}_{\mathbf{k}}^{-}\widetilde{S}_{-\mathbf{k}}^{+}\rangle\end{pmatrix}\right]_{(\alpha,\beta)_{\mu\nu}} \\ =& \sum_{\mu,\nu}\left[\mathsf{S}_{\mathbf{k}}\mathsf{T}_{\mathbf{k}}\mathsf{N}_{\mathbf{k}}\mathsf{T}_{\mathbf{k}}^{\dagger}\mathsf{S}_{\mathbf{k}}^{\dagger}\right]_{(\alpha,\beta)_{\mu\nu}},\end{aligned}$$

where the matrix elements are evaluated in the rotated spin basis.

The quantity relevant to neutron scattering experiments is defined as:

$$\mathcal{S}(\mathbf{q}) \equiv \frac{1}{N}\sum_{\substack{\alpha,\beta\\ I,J}}\Big(\delta_{\alpha\beta}-\frac{q_\alpha q_\beta}{q^2}\Big)e^{i\mathbf{q}\cdot(\mathbf{r}_I-\mathbf{r}_J)}\langle S_i^{\alpha}S_j^{\beta}\rangle = \sum_{\alpha,\beta}\Big(\delta_{\alpha\beta}-\frac{q_\alpha q_\beta}{q^2}\Big)\mathcal{S}^{\alpha\beta}(\mathbf{q}), \tag{128}$$

This formulation respects the directional constraints intrinsic to neutron scattering physics.

Dynamic structure factor

The dynamic structure factor is defined as

$$\mathcal{S}^{\alpha\beta}(\mathbf{k},\omega) = \frac{1}{2\pi}\int_{-\infty}^{\infty} dt\, e^{i\omega t}\mathcal{C}^{\alpha\beta}(\mathbf{k},t). \tag{129}$$

From the LSWT, we consider

$$\begin{aligned}\mathcal{S}_{\mu\nu}^{\alpha\beta}(\mathbf{k},\omega) =& \sum_{(m,n)_{\mu,\nu}}\int_{-\infty}^{\infty} dt\,[\mathsf{C}^{\alpha\beta}(\mathbf{k},t)]_{m,n}e^{i\omega t} = \sum_{(m,n)_{\mu,\nu}}\left[\int_{-\infty}^{\infty} dt\, e^{i\omega t}\mathsf{U}^{\alpha}\mathsf{S}_{\mathbf{k}}\mathsf{T}_{\mathbf{k}}\mathsf{N}_{\mathbf{k}}(t)\mathsf{T}_{\mathbf{k}}^{\dagger}\mathsf{S}_{\mathbf{k}}^{\dagger}\mathsf{U}^{\beta}\right]_{m,n} \\ =& \sum_{(m,n)_{\mu,\nu}}\left[\mathsf{R}_{\mathbf{k}}^{\alpha}\mathsf{T}_{\mathbf{k}}\left(\int_{-\infty}^{\infty} dt\, e^{i\omega t}\mathsf{N}_{\mathbf{k}}(t)\right)\mathsf{T}_{\mathbf{k}}^{\dagger}[\mathsf{R}_{\mathbf{k}}^{\beta}]^{\dagger}\right]_{m,n}.\end{aligned}$$

However, the integral yields δ-function. So we introduce the decay factor $\eta > 0$ so that $\omega \to \omega + i\eta$:

$$g_{\mathbf{k}}(\omega;\eta) \equiv \int_0^{\infty} dt\, e^{-[\eta+i(E_{\mathbf{k}}-\omega)]t} = \frac{1}{\eta+i(E_{\mathbf{k}}-\omega)} \tag{130}$$

Using this notation, we have

$$\mathcal{F}_{\mathbf{k}}(\omega;\eta) = \int_{-\infty}^{\infty} dt\, e^{-iE_{\mathbf{k}}t}e^{i\omega t-\eta|t|} = \int_0^{\infty} dt\,\left(e^{-[\eta+i(E_{\mathbf{k}}-\omega)]t}+e^{-[\eta-i(E_{\mathbf{k}}-\omega)]t}\right) = g_{\mathbf{k}}(\omega;\eta)+g_{\mathbf{k}}^{*}(\omega;\eta) \tag{131}$$

$$=2\mathrm{Re}[g_{\mathbf{k}}(\omega;\eta)] = \frac{2\eta}{\eta^2+(E_{\mathbf{k}}-\omega)^2} \tag{132}$$

where $\mathcal{F}_{\mathbf{k}}(\omega;\eta)$ is Lorentzian and its FWHM is 2η. **[REVIEW: A14: $\mathcal{F}_k$ is defined for scalar e^{-iEt} but $N_k(t)$ has different time dependence in upper-left and lower-right blocks. Make the connection explicit.]**

$$\begin{aligned}\int_{-\infty}^{\infty} dt\, e^{i\omega t-\eta|t|}\mathsf{N}_{\mathbf{k}}(t) =& \int_{-\infty}^{\infty} dt\, e^{i\omega t-\eta|t|}\begin{pmatrix}(\mathsf{I}+\mathsf{n}_{\mathbf{k}})e^{-iE_{\mathbf{k},\mu}t} & 0 \\ 0 & \mathsf{n}_{-\mathbf{k}}e^{+iE_{-\mathbf{k},\mu}t}\end{pmatrix} \\ =& \begin{pmatrix}(\mathsf{I}+\mathsf{n}_{\mathbf{k}})\int_{-\infty}^{\infty} dt\, e^{i\omega t-\eta|t|}e^{-iE_{\mathbf{k},\mu}t} & 0 \\ 0 & \mathsf{n}_{-\mathbf{k}}\int_{-\infty}^{\infty} dt\, e^{i\omega t-\eta|t|}e^{+iE_{-\mathbf{k},\mu}t}\end{pmatrix} \\ =& \begin{pmatrix}(\mathsf{I}+\mathsf{n}_{\mathbf{k}})\mathcal{F}_{\mathbf{k}}(\omega;\eta) & 0 \\ 0 & \mathsf{n}_{-\mathbf{k}}\mathcal{F}_{-\mathbf{k}}(-\omega;\eta)\end{pmatrix}\end{aligned} \tag{133}$$

Thus, in LSWT, the dynamical spin structure factor can be computed from the matrix multiplication:

$$\mathcal{S}^{\alpha\beta}_{\mu\nu}(\mathbf{k},\omega) = \sum_{(m,n)_{\mu,\nu}} \left[\mathsf{R}^{\alpha}_{\mathbf{k}}\mathsf{T}_{\mathbf{k}}\mathcal{F}[\mathsf{N}_{\mathbf{k}}](\omega;\eta)\mathsf{T}^{\dagger}_{\mathbf{k}}[\mathsf{R}^{\beta}_{\mathbf{k}}]^{\dagger}\right]_{m,n}. \tag{134}$$

Here, the fourier transformation of diagonal matrix $\mathsf{N}_{\mathbf{k}}(t)$ is given by

$$\mathcal{F}[\mathsf{N}_{\mathbf{k}}](\omega;\eta) \equiv \begin{pmatrix} (\mathsf{I}+\mathsf{n}_{\mathbf{k}})\mathcal{F}_{\mathbf{k}}(\omega;\eta) & 0 \\ 0 & \mathsf{n}_{-\mathbf{k}}\mathcal{F}_{-\mathbf{k}}(-\omega;\eta) \end{pmatrix} \tag{135}$$

C. Spectral Function

In this section, we compute the spectral function within Linear Spin-Wave Theory (LSWT):

$$\mathcal{A}^{\alpha\beta}(\mathbf{k},\omega) = -\frac{1}{\pi}\mathrm{Im}[G^{\alpha\beta}_{\mathrm{R}}(\mathbf{k},\omega)], \tag{136}$$

where $G^{\alpha\beta}_{\mathrm{R}}(\mathbf{k},\omega)$ is the retarded Green's function, defined as

$$G^{\alpha\beta}_{\mathrm{R}}(\mathbf{k},\omega) \equiv \frac{1}{i}\int_{-\infty}^{\infty} dt\, e^{i\omega t}\theta(t)\big\langle[S^{\alpha}_{\mathbf{k}}(t), S^{\beta}_{-\mathbf{k}}(0)]\big\rangle. \tag{137}$$

The ratared Green's function can be rewritten in terms of the correlation function

$$G^{\alpha\beta}_{\mathrm{R}}(\mathbf{k},\omega) = \frac{1}{i}\int_{0}^{\infty} dt\left[\big\langle S^{\alpha}_{\mathbf{k}}(t)S^{\beta}_{-\mathbf{k}}(0)\big\rangle - \big\langle S^{\beta}_{-\mathbf{k}}(0)S^{\alpha}_{\mathbf{k}}(t)\big\rangle\right]e^{i\omega t} = \frac{1}{i}\int_{0}^{\infty} dt\left[\mathcal{C}^{\alpha\beta}(\mathbf{k},t) - \mathcal{C}^{\beta\alpha}(-\mathbf{k},-t)\right]e^{i\omega t}. \tag{138}$$

For spin components $\alpha,\beta = x,y,z$, the retarded Green's function is expressed as the Fourier transform of the imaginary part of the correlation function: **[REVIEW: A15: Formula $G_R = 2\int \mathrm{Im}[C]e^{i\omega t}dt$ holds only for $\alpha,\beta = x,y,z$. The $\pm,0$ case needs $G_R = (1/i)\int[C^{\alpha\beta} - C^{\bar\beta\bar\alpha}(-k,-t)]e^{i\omega t}dt$. State explicitly if needed.]**

$$G^{\alpha\beta}_{\mathrm{R}}(\mathbf{k},\omega) = 2\int_{0}^{\infty} dt\,\mathrm{Im}[\mathcal{C}^{\alpha\beta}(\mathbf{k},t)]e^{i\omega t}. \tag{139}$$

The spectral function is then obtained as

$$\begin{aligned}
\mathcal{A}^{\alpha\beta}(\mathbf{k},\omega) &= -\frac{2}{\pi}\int_{0}^{\infty} dt\,\mathrm{Im}\big[\mathrm{Im}\big[\mathcal{C}^{\alpha\beta}(\mathbf{k},t)\big]e^{i\omega t}\big] \\
&= -\frac{2}{\pi}\int_{0}^{\infty} dt\,\mathrm{Im}\big[\mathcal{C}^{\alpha\beta}(\mathbf{k},t)\sin(\omega t)\big] \\
&= -\frac{2}{\pi}\mathrm{Im}\sum_{m,n}\left[\mathsf{U}^{\alpha}\mathsf{S}_{\mathbf{k}}\mathsf{T}_{\mathbf{k}}\left(\int_{0}^{\infty} dt\begin{pmatrix}(\mathsf{I}+\mathsf{n}_{\mathbf{k}})e^{-i\mathsf{E}_{\mathbf{k}}t} & 0 \\ 0 & \mathsf{n}_{-\mathbf{k}}e^{i\mathsf{E}_{-\mathbf{k}}t}\end{pmatrix}\sin(\omega t)\right)\mathsf{T}^{\dagger}_{\mathbf{k}}\bar{\mathsf{S}}_{\mathbf{k}}\bar{\mathsf{U}}^{\beta}\right]_{mn} \\
&= -\frac{2}{\pi}\mathrm{Im}\sum_{m,n}\left[\mathsf{U}^{\alpha}\mathsf{S}_{\mathbf{k}}\mathsf{T}_{\mathbf{k}}\begin{pmatrix}(\mathsf{I}+\mathsf{n}_{\mathbf{k}})\mathcal{G}_{\mathbf{k}}(\omega;\eta) & 0 \\ 0 & \mathsf{n}_{-\mathbf{k}}\mathcal{G}^{*}_{-\mathbf{k}}(\omega;\eta)\end{pmatrix}\mathsf{T}^{\dagger}_{\mathbf{k}}\bar{\mathsf{S}}_{\mathbf{k}}\bar{\mathsf{U}}^{\beta}\right]_{m,n}
\end{aligned} \tag{140}$$

where the function $\mathcal{G}_{\mathbf{k}}(\omega;\eta)$ is defined as

$$\begin{aligned}
\mathcal{G}_{\mathbf{k}}(\omega;\eta) \equiv& \int_{0}^{\infty} dt\, e^{-\mathrm{i}E_{\mathbf{k}}t}\sin(\omega t)e^{-\eta|t|} = \int_{0}^{\infty} dt\, e^{-\mathrm{i}E_{\mathbf{k}}t}\left(\frac{e^{\mathrm{i}\omega t} - e^{-\mathrm{i}\omega t}}{2\mathrm{i}}\right)e^{-\eta t} \\
=&\frac{1}{2\mathrm{i}}\left(\int_{0}^{\infty} dt\, e^{-(\eta+\mathrm{i}(E_{\mathbf{k}}-\omega))t} - \int_{0}^{\infty} dt\, e^{-(\eta+\mathrm{i}(E_{\mathbf{k}}+\omega))t}\right) = \frac{1}{2\mathrm{i}}\Big(\frac{1}{\eta+\mathrm{i}(E_{\mathbf{k}}-\omega)} - \frac{1}{\eta+\mathrm{i}(E_{\mathbf{k}}+\omega)}\Big) \\
=&\frac{1}{2i}\big(g_{\eta}(\mathbf{k},\omega) + g_{\eta}(\mathbf{k},-\omega)\big).
\end{aligned}$$

Thus, the spectral function in LSWT can be computed from the product of the matrices.

$$\mathcal{A}^{\alpha\beta}(\mathbf{k},\omega) = -\frac{2}{\pi}\mathrm{Im}\sum_{m,n}\left[\mathsf{U}^{\alpha}\mathsf{S}_{\mathbf{k}}\mathsf{T}_{\mathbf{k}}\mathcal{G}[\mathsf{N}_{\mathbf{k}}](\omega;\eta)\mathsf{T}^{\dagger}_{\mathbf{k}}\bar{\mathsf{S}}_{\mathbf{k}}\bar{\mathsf{U}}^{\beta}\right]_{m,n}, \tag{141}$$

where the matrix $\mathcal{G}[\mathsf{N}_{\mathbf{k}}](\omega;\eta)$ is defined as

$$\mathcal{G}[\mathsf{N}_{\mathbf{k}}](\omega;\eta) = \begin{pmatrix}(\mathsf{I}+\mathsf{n}_{\mathbf{k}})\mathcal{G}_{\mathbf{k}}(\omega;\eta) & 0 \\ 0 & \mathsf{n}_{-\mathbf{k}}\mathcal{G}^{*}_{-\mathbf{k}}(\omega;\eta).\end{pmatrix}. \tag{142}$$

V. SKYRMIONS, TOPOLOGICAL MAGNONS, AND HALL EFFECTS

A. Skyrmion number

The skyrmion number itself does not require the LSWT, but introducing it here would be useful for a later purpose. **[REVIEW: B9: Broken/orphaned text from incomplete edit ('ho in the lattice [cite] However,'). Remove or complete this sentence fragment.]** ho in the lattice [**?**] However, The skyrmion number $Q_{\rm sk}$ quantifies the topological charge of a magnetic skyrmion in a lattice, computed as the sum of solid angles over triangular plaquettes ($\triangle$):

$$Q_{\rm skyr} = \frac{1}{4\pi} \sum_{\triangle} \chi_{\triangle}. \tag{143}$$

For example, when the subsystem has four magnetic sublattices ($\mu = a, b, c, d$), the summation runs over the triangles abc, acd, abd, bcd. Here, $\chi_{\triangle}$ is the solid angle subtended by normalized magnetization vectors $\mathbf{m}_i$, $\mathbf{m}_j$, and $\mathbf{m}_k$ at the vertices of a plaquette, where $\mathbf{m}_i = \mathbf{S}_i/S_i$ with $|\mathbf{m}_i| = 1$. The solid angle is given by:

$$\chi_{\triangle} = 2 \arctan \left(\frac{|\mathbf{m}_i \cdot (\mathbf{m}_j \times \mathbf{m}_k)|}{1 + \mathbf{m}_i \cdot \mathbf{m}_j + \mathbf{m}_j \cdot \mathbf{m}_k + \mathbf{m}_k \cdot \mathbf{m}_i} \right). \tag{144}$$

B. Chern Number

The Chern number of the nth band, C_n, is defined as the integral of the Berry curvature $\Omega_{n,\mathbf{k}}$ over the first Brillouin zone (FBZ):

$$C_n = \frac{1}{2\pi} \int_{\rm FBZ} \Omega_{n,\mathbf{k}}\, d^2\mathbf{k} \approx \frac{1}{2\pi} \sum_{\mathbf{k}\in{\rm FBZ}} \Omega_{n,\mathbf{k}} \Delta k_x \Delta k_y. \tag{145}$$

The Berry curvature for the nth band is calculated as **[REVIEW: A6: Sum runs $m = 1$ to $2N$. Bands $n = 1..N$ are physical, $n = N + 1..2N$ redundant. Denominator involves positive-negative energy differences. Consistent with Shindou 2013 but should be stated explicitly.]**

$$\Omega^{\mu\nu}_{n,\mathbf{k}} = -2\, {\rm Im} \sum_{\substack{m=1\\ m\neq n}}^{2N} \frac{\left[\mathsf{J}\mathsf{T}^\dagger_{\mathbf{k}}(\partial_\mu \mathsf{H}_{\mathbf{k}})\mathsf{T}_{\mathbf{k}}\right]_{nm} \left[\mathsf{J}\mathsf{T}^\dagger_{\mathbf{k}}(\partial_\nu \mathsf{H}_{\mathbf{k}})\mathsf{T}_{\mathbf{k}}\right]_{mn}}{\left((\mathsf{J}\mathsf{E}_{\mathbf{k}})_{nn} - (\mathsf{J}\mathsf{E}_{\mathbf{k}})_{mm}\right)^2}, \tag{146}$$

where J, $\mathsf{H}_{\mathbf{k}}$, $\mathsf{T}_{\mathbf{k}}$, and $\mathsf{E}_{\mathbf{k}}$ are $2N \times 2N$ matrices, defined earlier

$$\mathsf{J} = \begin{pmatrix} \mathsf{I} & 0 \\ 0 & -\mathsf{I} \end{pmatrix}, \qquad \text{and} \qquad \mathsf{T}^\dagger_{\mathbf{k}} \mathsf{H}_{\mathbf{k}} \mathsf{T}_{\mathbf{k}} = \mathsf{E}_{\mathbf{k}}, \tag{147}$$

with $\mathsf{T}_{\mathbf{k}}$ diagonalizing the Hamiltonian matrix $\mathsf{H}_{\mathbf{k}}$ to yield the energy matrix $\mathsf{E}_{\mathbf{k}}$.

C. Thermal Hall Conductance

The thermal Hall conductivity (THC), while it's hard to observe in experiments, is related to the topology of magnon bands [**?**]. It is computed using the Berry curvature as

$$\kappa_{xy} = -\frac{k_B^2 T}{V} \sum_{\mathbf{k}\in{\rm FBZ}} \sum_{n=1}^{N} \left\{ c_2[g(\varepsilon_{n,\mathbf{k}})] - \frac{\pi^2}{3} \right\} \Omega_{n,\mathbf{k}}, \tag{148}$$

[REVIEW: A7: THC sums $n = 1$ to N (physical bands only) but Berry curvature is defined in $2N$ space. Verify consistency.] **[REVIEW: A16: $\varepsilon_{n,k}$ appears suddenly; rest of document uses $E_{k,n}$. Unify notation.]** where V is the system volume, $g(\varepsilon_{n,\mathbf{k}}) = [e^{\varepsilon_{n,\mathbf{k}}/k_B T} - 1]^{-1}$ is the Bose-Einstein distribution, and $c_2(x)$ is the Spence function, defined as

$$c_2(x) = \int_0^x dt \left(\ln \frac{1+t}{t} \right)^2 = (1+x) \left(\ln \frac{1+x}{x} \right)^2 - (\ln x)^2 - 2{\rm Li}_2(-x), \quad \text{with} \quad {\rm Li}_2(z) = -\int_0^z \frac{\ln(1-t)}{t}\, dt. \tag{149}$$

VI. EXAMPLE: SOLVING SPIN SYSTEM WITH LINEAR SPIN WAVE THEORY

In this example, we solve a quadratic boson Hamiltonian. Consider the following Hamiltonian in momentum space, rewritten in Nambu form:

$$H = \sum_{\mathbf{k}} A_{\mathbf{k}} a_{\mathbf{k}}^\dagger a_{\mathbf{k}} + \frac{1}{2}\left(B_{\mathbf{k}} a_{\mathbf{k}}^\dagger a_{-\mathbf{k}}^\dagger + B_{\mathbf{k}}^* a_{\mathbf{k}} a_{-\mathbf{k}}\right) = \frac{1}{2}\sum_{\mathbf{k}} \begin{pmatrix} a_{\mathbf{k}}^\dagger & a_{-\mathbf{k}} \end{pmatrix} \begin{pmatrix} A_{\mathbf{k}} & B_{\mathbf{k}} \\ B_{\mathbf{k}}^* & A_{\mathbf{k}} \end{pmatrix} \begin{pmatrix} a_{\mathbf{k}} \\ a_{-\mathbf{k}}^\dagger \end{pmatrix}, \tag{150}$$

where $B_{\mathbf{k}}$ is real without loss of generality. This assumption holds because we can redefine the bosonic operator with a phase factor $a_{\mathbf{k}} \to a_{\mathbf{k}} e^{-i\phi}$, where $B_{\mathbf{k}} = e^{i\phi}|B_{\mathbf{k}}|$, making $B_{\mathbf{k}}$ real by choosing an appropriate ϕ. Note that the condition to be Hamiltonian density positive is

$$\lambda(\mathsf{H}_{\mathbf{k}}) = A_{\mathbf{k}} \pm B_{\mathbf{k}} \geq 0, \qquad \to \qquad |A_{\mathbf{k}}| \geq |B_{\mathbf{k}}|.$$

This condition corresponds to the positivity of the the magnon energy $\omega_{\mathbf{k}} = \sqrt{A_{\mathbf{k}}^2 - B_{\mathbf{k}}^2}$.

To diagonalize the Hamiltonian, we introduce the Bogoliubov transformation:

$$\begin{pmatrix} a_{\mathbf{k}} \\ a_{-\mathbf{k}}^\dagger \end{pmatrix} = \begin{pmatrix} \cosh\theta_{\mathbf{k}} & \sinh\theta_{-\mathbf{k}} \\ \sinh\theta_{\mathbf{k}} & \cosh\theta_{-\mathbf{k}} \end{pmatrix} \begin{pmatrix} \alpha_{\mathbf{k}} \\ \alpha_{-\mathbf{k}}^\dagger \end{pmatrix} \tag{151}$$

where $\theta_{\mathbf{k}}$ is real and even under $\mathbf{k} \to -\mathbf{k}$, so $\theta_{\mathbf{k}} = \theta_{-\mathbf{k}}$. The bosonic commutation relations $[a_{\mathbf{k}}, a_{\mathbf{k}}^\dagger] = 1$ and $[\alpha_{\mathbf{k}}, \alpha_{\mathbf{k}}^\dagger] = 1$ must be preserved. Define the transformation matrix as:

$$\mathsf{T}_{\mathbf{k}} = \begin{pmatrix} \cosh\theta_{\mathbf{k}} & \sinh\theta_{\mathbf{k}} \\ \sinh\theta_{\mathbf{k}} & \cosh\theta_{\mathbf{k}} \end{pmatrix} \tag{152}$$

Check the commutation relation for the new operators:

$$\begin{aligned} [\alpha_{\mathbf{k}}, \alpha_{\mathbf{k}}^\dagger] &= [\cosh\theta_{\mathbf{k}} a_{\mathbf{k}} + \sinh\theta_{\mathbf{k}} a_{-\mathbf{k}}^\dagger, \cosh\theta_{\mathbf{k}} a_{\mathbf{k}}^\dagger + \sinh\theta_{\mathbf{k}} a_{-\mathbf{k}}] && (153) \\ &= \cosh^2\theta_{\mathbf{k}} [a_{\mathbf{k}}, a_{\mathbf{k}}^\dagger] + \sinh^2\theta_{\mathbf{k}} [a_{-\mathbf{k}}^\dagger, a_{-\mathbf{k}}] && (154) \\ &= \cosh^2\theta_{\mathbf{k}} \cdot 1 + \sinh^2\theta_{\mathbf{k}} \cdot (-1) && (155) \\ &= \cosh^2\theta_{\mathbf{k}} - \sinh^2\theta_{\mathbf{k}} = 1 && (156) \end{aligned}$$

This holds due to the hyperbolic identity, confirming that the transformation is canonical.

Diagonalizing the Hamiltonian

[REVIEW: C15: Unnumbered subsection won't appear in TOC. Change to \subsection or keep intentionally.]

Substitute the transformation into the Hamiltonian. First, the conjugate transformation is:

$$\begin{pmatrix} a_{\mathbf{k}}^\dagger & a_{-\mathbf{k}} \end{pmatrix} = \begin{pmatrix} \alpha_{\mathbf{k}}^\dagger & \alpha_{-\mathbf{k}} \end{pmatrix} \begin{pmatrix} \cosh\theta_{\mathbf{k}} & \sinh\theta_{\mathbf{k}} \\ \sinh\theta_{\mathbf{k}} & \cosh\theta_{\mathbf{k}} \end{pmatrix} \tag{157}$$

Thus, the Hamiltonian becomes:

$$H = \frac{1}{2}\sum_{\mathbf{k}} \begin{pmatrix} \alpha_{\mathbf{k}}^\dagger & \alpha_{-\mathbf{k}} \end{pmatrix} T_{\mathbf{k}} \begin{pmatrix} A_{\mathbf{k}} & B_{\mathbf{k}} \\ B_{\mathbf{k}}^* & A_{\mathbf{k}} \end{pmatrix} T_{\mathbf{k}} \begin{pmatrix} \alpha_{\mathbf{k}} \\ \alpha_{-\mathbf{k}}^\dagger \end{pmatrix} \tag{158}$$

Define the matrix to be diagonalized as:

$$\mathsf{E}_{\mathbf{k}} = \mathsf{T}_{\mathbf{k}}^\dagger \begin{pmatrix} A_{\mathbf{k}} & B_{\mathbf{k}} \\ B_{\mathbf{k}}^* & A_{\mathbf{k}} \end{pmatrix} \mathsf{T}_{\mathbf{k}} = \begin{pmatrix} \cosh\theta_{\mathbf{k}} & \sinh\theta_{\mathbf{k}} \\ \sinh\theta_{\mathbf{k}} & \cosh\theta_{\mathbf{k}} \end{pmatrix} \begin{pmatrix} A_{\mathbf{k}} & B_{\mathbf{k}} \\ B_{\mathbf{k}}^* & A_{\mathbf{k}} \end{pmatrix} \begin{pmatrix} \cosh\theta_{\mathbf{k}} & \sinh\theta_{\mathbf{k}} \\ \sinh\theta_{\mathbf{k}} & \cosh\theta_{\mathbf{k}} \end{pmatrix} \tag{159}$$

Since $B_{\mathbf{k}}$ is real, $B_{\mathbf{k}}^* = B_{\mathbf{k}}$. Compute $M_{\mathbf{k}}$:

$$\mathsf{E}_{\mathbf{k}} = \begin{pmatrix} A_{\mathbf{k}}\cosh 2\theta_{\mathbf{k}} + B_{\mathbf{k}}\sinh 2\theta_{\mathbf{k}} & A_{\mathbf{k}}\sinh 2\theta_{\mathbf{k}} + B_{\mathbf{k}}\cosh 2\theta_{\mathbf{k}} \\ A_{\mathbf{k}}\sinh 2\theta_{\mathbf{k}} + B_{\mathbf{k}}\cosh 2\theta_{\mathbf{k}} & A_{\mathbf{k}}\cosh 2\theta_{\mathbf{k}} + B_{\mathbf{k}}\sinh 2\theta_{\mathbf{k}} \end{pmatrix} = \begin{pmatrix} \omega_{\mathbf{k}} & 0 \\ 0 & \omega_{\mathbf{k}} \end{pmatrix} \tag{160}$$

Thus, we have the diagonalization condition

$$[\mathsf{E}_{\mathbf{k}}]_{12} = A_{\mathbf{k}} \sinh 2\theta_{\mathbf{k}} + B_{\mathbf{k}} \cosh 2\theta_{\mathbf{k}} = 0, \quad \rightarrow \quad \tanh 2\theta_{\mathbf{k}} = -\frac{B_{\mathbf{k}}}{A_{\mathbf{k}}} \tag{161}$$

The diagonal element $M_{\mathbf{k}}^{(11)}$ gives the eigenenergy:

$$\omega_{\mathbf{k}} = A_{\mathbf{k}} \cosh 2\theta_{\mathbf{k}} + B_{\mathbf{k}} \sinh 2\theta_{\mathbf{k}} = A_{\mathbf{k}} \cdot \frac{A_{\mathbf{k}}}{\sqrt{A_{\mathbf{k}}^2 - B_{\mathbf{k}}^2}} + B_{\mathbf{k}} \cdot \left(-\frac{B_{\mathbf{k}}}{\sqrt{A_{\mathbf{k}}^2 - B_{\mathbf{k}}^2}} \right) = \sqrt{A_{\mathbf{k}}^2 - B_{\mathbf{k}}^2}. \tag{162}$$

To express this in terms of $A_{\mathbf{k}}$ and $B_{\mathbf{k}}$, use $\cosh 2\theta_{\mathbf{k}} = \frac{1}{\sqrt{1-\tanh^2 2\theta_{\mathbf{k}}}}$ and $\sinh 2\theta_{\mathbf{k}} = \tanh 2\theta_{\mathbf{k}} \cosh 2\theta_{\mathbf{k}}$:

$$\cosh 2\theta_{\mathbf{k}} = \frac{1}{\sqrt{1 - (-B_{\mathbf{k}}/A_{\mathbf{k}})^2}} = \frac{A_{\mathbf{k}}}{\omega_{\mathbf{k}}}, \qquad \sinh 2\theta_{\mathbf{k}} = -\frac{B_{\mathbf{k}}}{A_{\mathbf{k}}} \cdot \frac{A_{\mathbf{k}}}{\sqrt{A_{\mathbf{k}}^2 - B_{\mathbf{k}}^2}} = -\frac{B_{\mathbf{k}}}{\omega_{\mathbf{k}}}.$$

[REVIEW: B21: 'Number Expectation in Ground State' as \noindent\textbf{}. Promote to \subsection or use \subparagraph for consistency with document structure.] **Number Expectation in Ground State:**

$$\begin{aligned}
\langle n_{\mathbf{k}} \rangle =& \langle a_{\mathbf{k}}^\dagger a_{\mathbf{k}} \rangle = \langle (\cosh\theta_{\mathbf{k}} \alpha_{\mathbf{k}}^\dagger + \sinh\theta_{\mathbf{k}} \alpha_{-\mathbf{k}})(\cosh\theta_{\mathbf{k}} \alpha_{\mathbf{k}} + \sinh\theta_{\mathbf{k}} \alpha_{-\mathbf{k}}^\dagger)\rangle & (163)\\
=& \cosh^2\theta_{\mathbf{k}} \langle \alpha_{\mathbf{k}}^\dagger \alpha_{\mathbf{k}} \rangle + \sinh^2\theta_{\mathbf{k}} \langle \alpha_{-\mathbf{k}} \alpha_{-\mathbf{k}}^\dagger \rangle \ + \sinh\theta_{\mathbf{k}} \cosh\theta_{\mathbf{k}} \big(\langle \alpha_{-\mathbf{k}}^\dagger \alpha_{\mathbf{k}}^\dagger \rangle + \langle \alpha_{\mathbf{k}} \alpha_{-\mathbf{k}} \rangle \big) & (164)\\
=& \sinh^2\theta_{\mathbf{k}} = \frac{\cosh 2\theta_{\mathbf{k}} - 1}{2} = \frac{A_{\mathbf{k}}/\omega_{\mathbf{k}} - 1}{2}. & (165)
\end{aligned}$$

- If $\omega_{\mathbf{k}} = \sqrt{A_{\mathbf{k}}^2 - |B_{\mathbf{k}}|^2} \rightarrow 0^+$, then the number

$$\langle a_k^\dagger a_k \rangle = |v_k|^2 = \frac{A_{\mathbf{k}}/\omega_{\mathbf{k}} - 1}{2} \approx \frac{A_{\mathbf{k}}}{\omega_{\mathbf{k}}} \rightarrow \infty \tag{166}$$

- If $B_{\mathbf{k}} = 0$:

$$\langle n_{\mathbf{k}} \rangle = 0 \tag{167}$$

- If $A_{\mathbf{k}} \rightarrow A_{\mathbf{k}} + \mu$, which corresponds to adding on-site chemical potential amount to $\mu \sum_{\mathbf{k}} a_{\mathbf{k}}^\dagger a_{\mathbf{k}} = \frac{\mu}{2} \sum_{\mathbf{k}} (a_{\mathbf{k}}^\dagger a_{\mathbf{k}} + a_{-\mathbf{k}} a_{-\mathbf{k}}^\dagger)$ is the quantum correction

$$\begin{aligned}
E_{\text{qu}}^{(\mu)} =& \frac{1}{2} \sum_{\mathbf{k}} \Big(\sum_{\mu} E_{\mathbf{k},\mu} - \mathrm{Tr}[\mathsf{A}_{\mathbf{k}}] \Big) = \frac{1}{2} \sum_{\mathbf{k}} \Big(((A_{\mathbf{k}} + \mu)^2 - B_{\mathbf{k}}^2)^{1/2} - (A_{\mathbf{k}} + \mu) \Big) \\
=& \frac{1}{2} \sum_{\mathbf{k}} \big((A_{\mathbf{k}}^2 - B_{\mathbf{k}})^{1/2} - A_{\mathbf{k}} \big) + \Big(\frac{A_{\mathbf{k}}}{\omega_{\mathbf{k}}} - 1 \Big) \mu + \frac{B_{\mathbf{k}}^2}{2\omega_{\mathbf{k}}^3} \mu^2 + \cdots \\
=& \sum_{\mathbf{k}} \frac{\omega_{\mathbf{k}}}{2} + \langle a_{\mathbf{k}}^\dagger a_{\mathbf{k}} \rangle \mu + \frac{\langle a_{\mathbf{k}}^\dagger a_{\mathbf{k}}^\dagger a_{\mathbf{k}} a_{\mathbf{k}} \rangle}{\omega_{\mathbf{k}}} \mu^2 + \cdots .
\end{aligned}$$

$$\begin{aligned}
\langle n_{\mathbf{k}}^{(\mu)} \rangle - \langle n_{\mathbf{k}}^{(0)} \rangle =& \frac{A_{\mathbf{k}} + \mu - \sqrt{(A_{\mathbf{k}} + \mu)^2 - B_{\mathbf{k}}^2}}{2\sqrt{(A_{\mathbf{k}} + \mu)^2 - B_{\mathbf{k}}^2}} - \frac{A_{\mathbf{k}} - \sqrt{(A_{\mathbf{k}})^2 - B_{\mathbf{k}}^2}}{2\sqrt{(A_{\mathbf{k}})^2 - B_{\mathbf{k}}^2}} \\
=& -\frac{B_{\mathbf{k}}^2}{2\omega_{\mathbf{k}}^3} \mu - \frac{3}{4} \frac{A_{\mathbf{k}} B_{\mathbf{k}}^2}{\omega_{\mathbf{k}}^5} \mu^2 + \cdots
\end{aligned}$$